

# Fringe fields for combined function magnets

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Science & Technology Facilities Council

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# Overview & warning

- Motivation
  - Why do we want to do this ?
- Dipoles
  - Very simplified model to inform what may happen for higher order multipoles
- Multipoles
  - Extension of dipole theory to multipoles ? (details at: <http://arxiv.org/abs/1404.1762>)
- Quadrupoles
  - Details of how the theory works for the second order multipole
- HL-LHC Inner triplet application
- EMMA application
- Extension to combined function magnets
- GPT implementation
- Possible future directions
- Conclusions

My idea is this: There are many slides so, in the interest of time, I shall go through them rather fast. However, you can refer back to the slides of this talk if you need more information. If there is something missing or which you do not understand, please do not hesitate to contact me and I shall do my best to answer all your questions.

Acknowledgements: colleagues in ASTeC and everyone who has given me feedback about what could be done with the theory and what should be explained better !

# Motivation / History

Hard edge models are not realistic when it comes to modelling magnets in an accelerator, in particular, a small accelerator such as EMMA where fringe fields dominate and overlap between magnets.

Fringe fields for dipoles are well understood and, often the form assumed is

$$B_y = \frac{B}{2\pi} \left[ \pi - \arctan \left( \frac{z}{g+y} \right) - \arctan \left( \frac{z}{g-y} \right) \right]$$

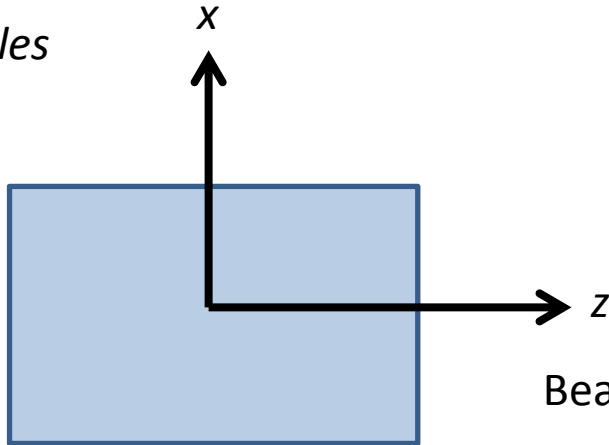
$$B_z = \frac{B}{4\pi} \left[ \ln(z^2 + (g+y)^2) - \ln(z^2 + (g-y)^2) \right].$$

The main problem with this expression is that, though it does indeed decay, it does so only at infinity (in  $z$ ) rather than earlier. This can create problems with space charge solvers where the fringe fields are truncated because, even if they look like zero, they are not. Therefore, if the field is truncated, this creates a discontinuity and, typically, the space charge algorithm dwells on this discontinuity and so takes a very long time to come up with an answer.

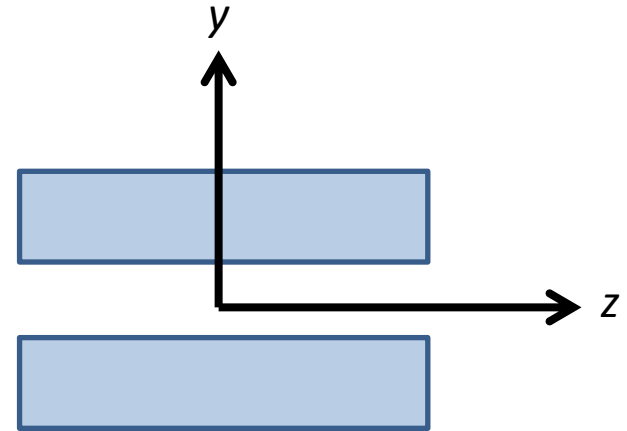
Fringe fields for quadrupoles and higher order multipoles are approximated via an expansion on-axis. This expansion is rather approximate when one goes considerably off-axis as is the case for example in the EMMA ns-FFAG or the HL-LHC inner triplet.

# Choice of coordinates

*Dipoles*

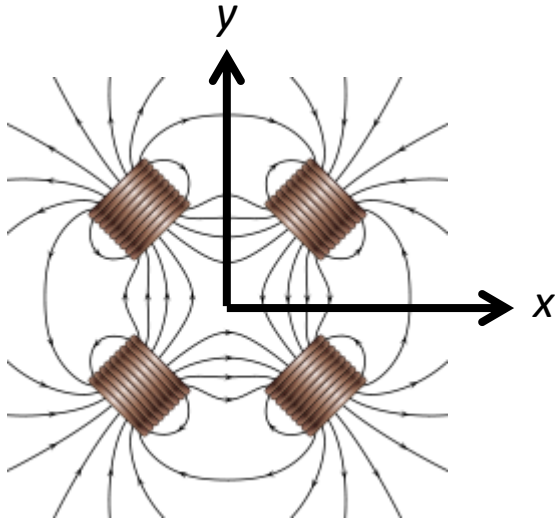


View from the top

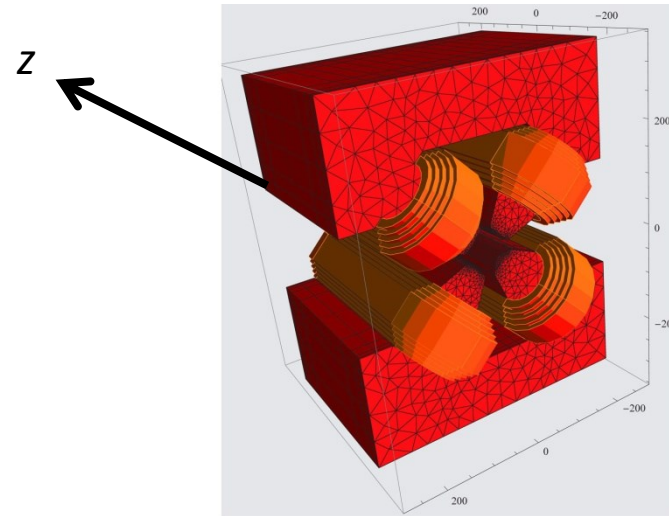


View from the side

With a similar set-up for quadrupoles and higher orders,  $z$  is the longitudinal coordinate



Transverse cross-section



# Fringe fields - dipoles (1)

Magnetostatic Maxwell  $\vec{\nabla} \times \vec{B} = \vec{\nabla} \cdot \vec{B} = 0. \quad \partial_y B_y + \partial_z B_z = \partial_y B_z - \partial_z B_y = 0$

$$\Delta_{y,z} B_y = \Delta_{y,z} B_z = 0 \quad (B_x = 0)$$

Engel-type decay  $\frac{1}{1 + \exp[E(z)]} \quad E(z) = a_0 + a_1 \left(\frac{z}{D}\right) + a_2 \left(\frac{z}{D}\right)^2 + \dots + a_5 \left(\frac{z}{D}\right)^5$

where the Engel coefficients ( $a_0, a_1, a_2, a_3, a_4, a_5$ ) influence the fall-off in the  $z$  direction.

There are many other (possibly better ?) decays possible – this method is always valid

Solution

$$\begin{aligned} B_y &= e(z + iy) + \bar{e}(z - iy) \\ B_z &= -ie(z + iy) + i\bar{e}(z - iy) \end{aligned} \quad \text{for any function } e(z+iy).$$

Particular case

$$E(z + iy) = z + iy$$

Gives

$$B_y = \frac{(1 + e^z \cos y)}{1 + 2e^z \cos y + e^{2z}}, \quad B_z = \frac{-e^z \sin y}{1 + 2e^z \cos y + e^{2z}}.$$

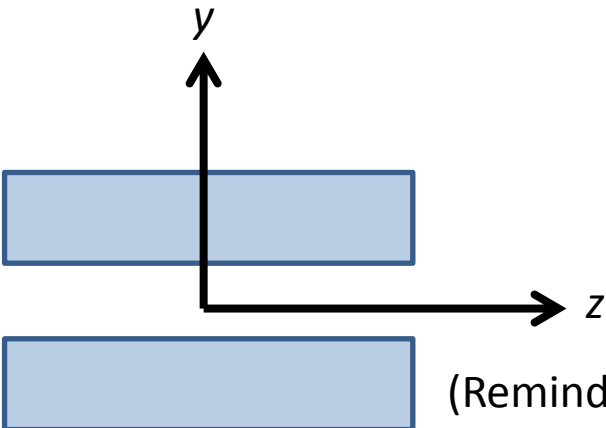
$B_z$  is required because we wish  $B_y$  to decay (along  $z$ ) while still satisfying Maxwell.

# Fringe fields – dipoles (2)

Singularities are necessary to have non-trivial solutions. These can be neglected as they lie outside the magnet and are not in the bending plane ( $x$ ) and the beam never sees them. There is an Infinite number of them (along  $z = 0$ ).

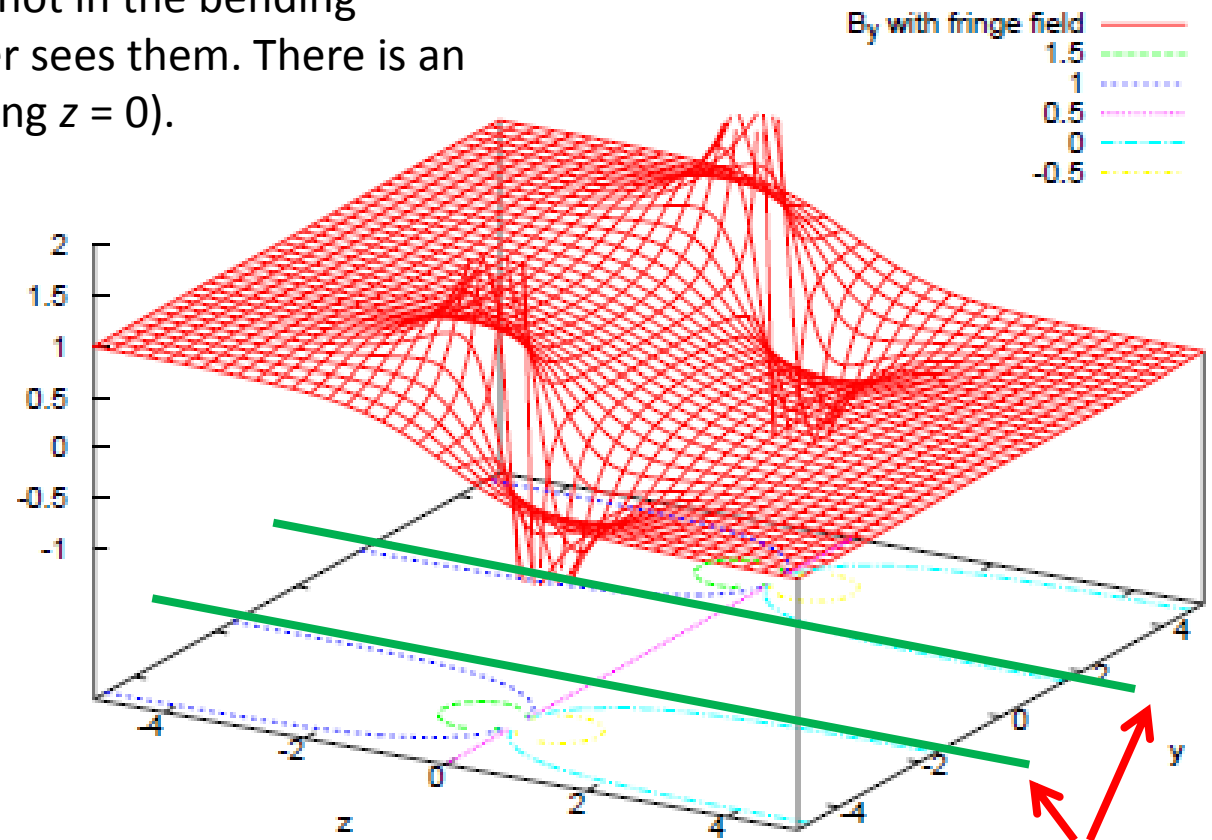
As you go up & down in the dipole ( $y$ ), the value of  $B_y$  increases as one goes away from the axis (in the fringe region) and reaches the pole.

$B_y$



(Reminder of dipole layout – view from side)

$B_y$  (dipoles) with fringe fields



Lines show edges ( $y$  only) of dipole which avoid singularities.

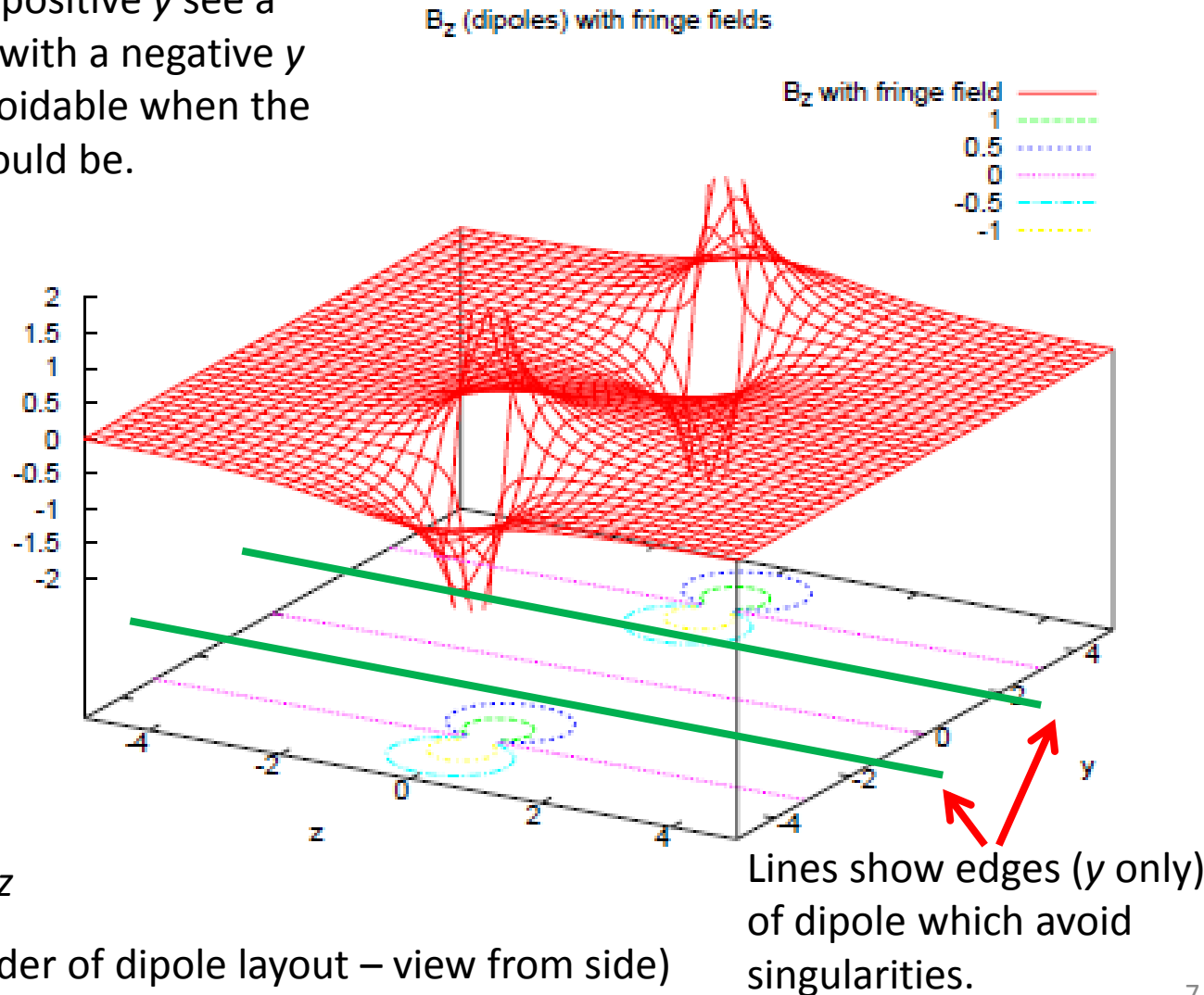
# Fringe fields – dipoles (3)

Particles on axis do not see a  $B_z$  field as is to be expected and particles with a positive  $y$  see a negative  $B_z$  whereas particles with a negative  $y$  see a positive  $B_z$ . This is unavoidable when the dipole is in the centre as it should be.

Both  $B_y$  and  $B_z$  are smooth in the  $(x)$  direction as expected.

Singularities may be thought of as where the current and coils are.

$B_z$



# Fringe fields - dipoles (4)

From static Maxwell we have  $\vec{\nabla} \times \vec{B} = \vec{\nabla} \cdot \vec{B} = 0$ .

Which implies the existence of the following two potentials

$$\vec{B} = \vec{\nabla} \varphi \qquad \vec{B} = \vec{\nabla} \times \vec{A} \qquad \text{respectively.}$$

For dipoles these two potentials may be written explicitly as

$$\varphi = y + i \ln \sqrt{\frac{1 + e^{z+iy}}{1 + e^{z-iy}}} + \text{const.}$$

with the only possible gauge freedom being a constant, and

$$A_x = z - \ln \sqrt{(1 + e^{z+iy})(1 + e^{z-iy})}$$

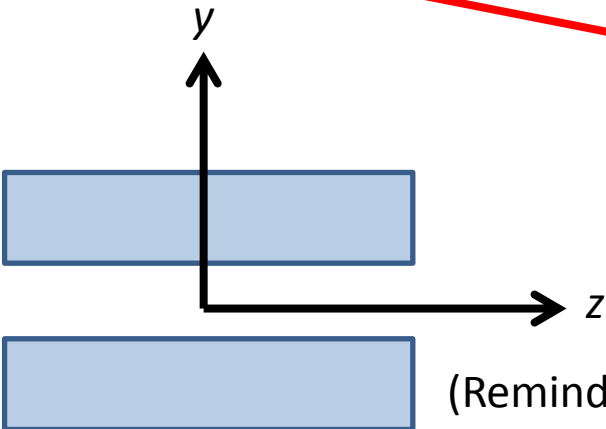
The gauge freedom in the vector potential above is extensive and other forms may be more appropriate, the one above is just one of the simplest with the other components set to zero. This should help considerably in finding a symplectic kick expression for the fringe field of a dipole which has already been done (Wu, Forest & Robin).



# Fringe fields - dipoles (5)

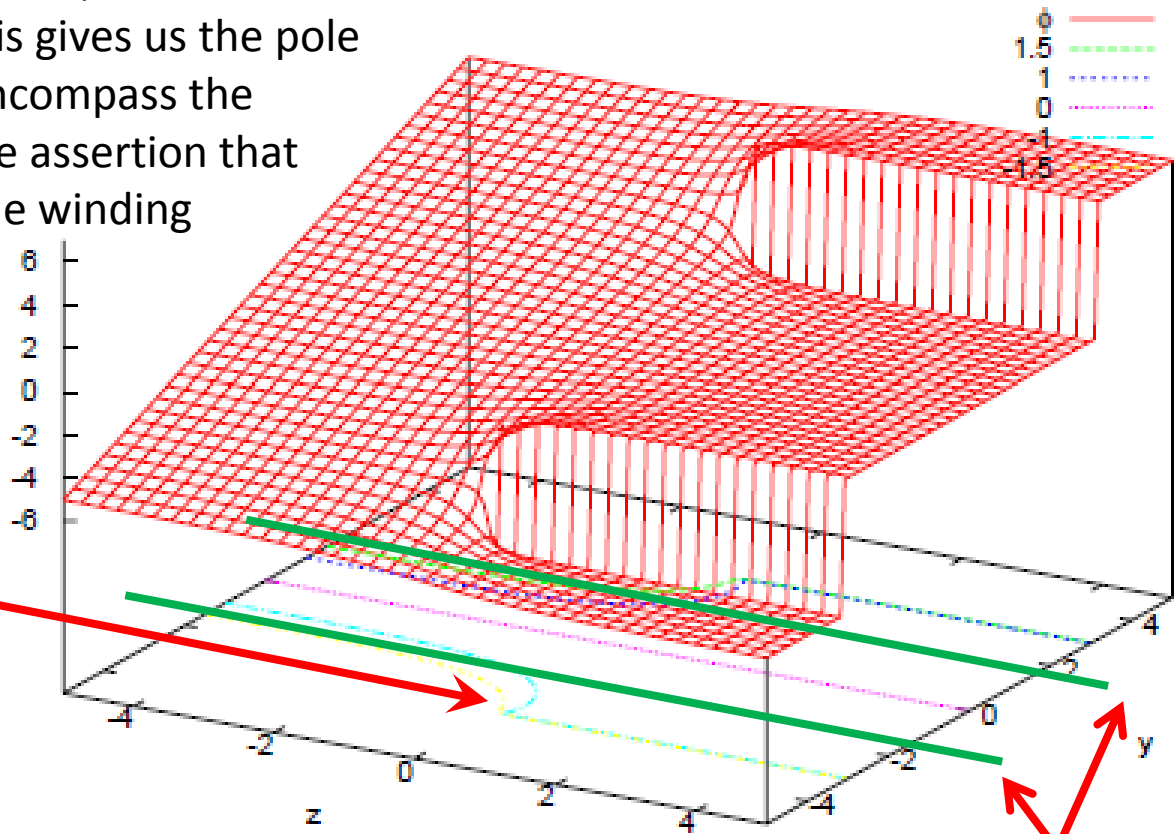
The scalar potential for the dipole field, being the integral of the field, shows the singularities as step functions. From the potential, we can look at the lines of constant  $\phi$  and this gives us the pole face geometry. The profiles encompass the singularities, agreeing with the assertion that these represent the coils of the winding of the dipole.

Contours showing possible pole-face profiles



(Reminder of dipole layout – view from side)

Potential for field with fringe fields



Lines show edges ( $y$  only) of dipole which avoid singularities.

# Fringe fields – multipoles

It is possible to show that the following is a solution of the 3D magneto-static equations (for the full derivation please see paper: <http://arxiv.org/abs/1404.1762>)

$$\begin{aligned}
 B_u &= -ibf(h+iz) + ibg(h-iz) & \text{for any functions } f(h+iz) \text{ and } g(h-iz) \text{ with} \\
 B_v &= -\frac{i}{b}f(h+iz) + \frac{i}{b}g(h-iz) & B_u = \frac{1}{\sqrt{2}}(B_x + iB_y) \quad B_v = \frac{1}{\sqrt{2}}(B_x - iB_y) \\
 B_z &= f(h+iz) + g(h-iz) & B'_z = B_z/\sqrt{2}, \quad z' = \sqrt{2}z
 \end{aligned}$$

where  $b$  is a constant and  $h$  is given by  $h(u, v) = \frac{1}{b}u + bv$ .

Note that  $B_u$  and  $B_v$  are proportional to each other so that it is not possible to describe a quadrupole with this solution. However, it is possible, due to the linearity of the equations, to add several of these solutions together as follows:

$$\begin{aligned}
 B_u &= \sum_{j=1}^m c_j \left[ -ib_j(h_j+iz)^n F_j(h_j+iz) + ib_j(h_j-iz)^n G_j(h_j-iz) \right] \\
 B_v &= \sum_{j=1}^m c_j \left[ -\frac{i}{b_j}(h_j+iz)^n F_j(h_j+iz) + \frac{i}{b_j}(h_j-iz)^n G_j(h_j-iz) \right] \\
 B_z &= \sum_{j=1}^m c_j \left[ (h_j+iz)^n F_j(h_j+iz) + (h_j-iz)^n G_j(h_j-iz) \right]
 \end{aligned}$$

# Fringe fields – quadrupoles (1)

where  $(h + iz)^n$  and  $(h - iz)^n$  represent the order of the multipole ( $n$ ) and  $F(h + iz)$  and  $G(h - iz)$  are multiplicative functions which we choose to give the correct behaviour for large  $z$  and which are 1 or -1 for large negative  $z$  ( $z = 0$  represents the region where the field is  $\frac{1}{2}$  the nominal value inside the magnet – this is just a property of the very simple Enge-style fall off that was chosen with  $a_1$  being the only non-zero coefficient).  $h_j = \frac{1}{b_j}u + b_jv$ ,  $m$  is a constant to be determined by how many terms are required in the sum and so are  $b_j$  and  $c_j$ . The constants  $b_j$  and  $c_j$  should satisfy simultaneous equations to ensure the correct behaviour of the field inside the magnet ( $z$  large and negative) with  $B_u \sim iv^n$ ,  $B_v \sim -iu^n$  and  $B_z = 0$ . The constants may be introduced due to the linearity of the Maxwell equations.

## Quadrupole equations

$$\begin{aligned}
 -ic_1b_1\left(\frac{u}{b_1} + b_1v + iz\right) - ic_1b_1\left(\frac{u}{b_1} + b_1v - iz\right) - ic_2b_2\left(\frac{u}{b_2} + b_2v + iz\right) - ic_2b_2\left(\frac{u}{b_2} + b_2v - iz\right) &= iv \\
 -i\frac{c_1}{b_1}\left(\frac{u}{b_1} + b_1v + iz\right) - i\frac{c_1}{b_1}\left(\frac{u}{b_1} + b_1v - iz\right) - i\frac{c_2}{b_2}\left(\frac{u}{b_2} + b_2v + iz\right) - i\frac{c_2}{b_2}\left(\frac{u}{b_2} + b_2v - iz\right) &= -iu \\
 -ic_1\left(\frac{u}{b_1} + b_1v + iz\right) - ic_1\left(\frac{u}{b_1} + b_1v - iz\right) - ic_2\left(\frac{u}{b_2} + b_2v + iz\right) - ic_2\left(\frac{u}{b_2} + b_2v - iz\right) &= 0
 \end{aligned}$$

# Fringe fields – quadrupoles (2)

$$2c_1b_1^2 + 2c_2b_2^2 = -1$$

$$2\frac{c_1}{b_1^2} + 2\frac{c_2}{b_2^2} = 1 \quad \text{giving the solution} \quad b_1 = \pm \frac{1}{b_2}, \quad c_1 = -c_2 = \frac{1}{2\left(b_2^2 - \frac{1}{b_2^2}\right)}.$$

$$c_1 + c_2 = 0$$

Note that  $m = 2$  for a quadrupole, more elementary solutions could have been included but this was not necessary. Also, because of the way the coefficient  $b$  appears we have that  $b \rightarrow -b$  and  $b \rightarrow 1/b$  gives  $B_x \rightarrow B_x$  and  $B_y \rightarrow -B_y$ . Further, we see that we cannot have  $b_1 = 0$  or 1 as this would lead to a singularity in either  $b_2$  or  $c_1$  and  $c_2$ .

The results are extendible to higher order multipoles in a straightforward way, however, the higher the order, the more three dimensional solutions discussed above need to be included.

This gives a solution for the fringe field of a quadrupole. However, this solution still has one remaining free constant, namely  $b$  ( $0 < b < 1$ ). This constant appears to be independent of the usual Enge coefficients ( $a_0, a_1, a_2, a_3, a_4, a_5$ ). However, this may not be entirely so and it shall be examined in more detail later on. For now,  $b$  appears to influence the rapidity of the fall-off in the transverse coordinates.

# Fringe fields – quadrupoles (4)

If we return to our usual magnetic field  $B_x, B_y$  and  $B_z$  in the usual coordinates  $(x, y, z)$ , with  $B_z$  and  $z$  were re-scaled:

$$B_x = \frac{1}{\sqrt{2}}(B_u + B_v) = \sum_{j=1}^2 \frac{c_j d_j}{2} [-2ih_j - \ln(1 + e^{\sqrt{2}z - ih_j}) + \ln(1 + e^{\sqrt{2}z + ih_j})]$$

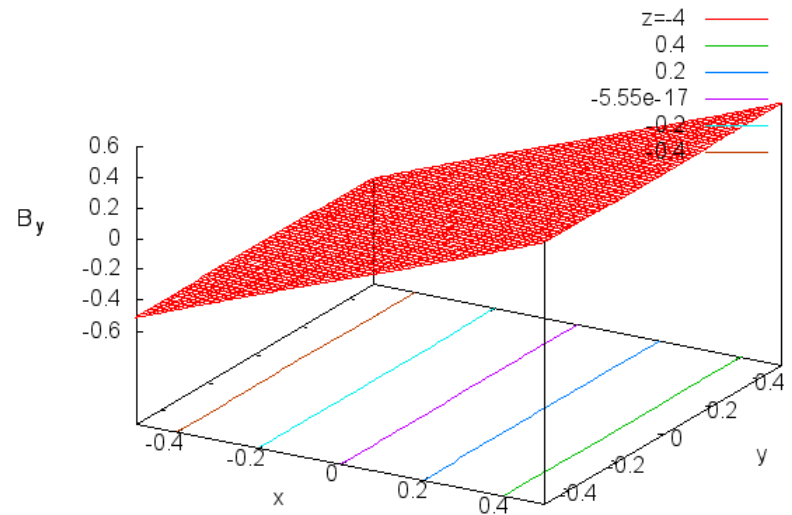
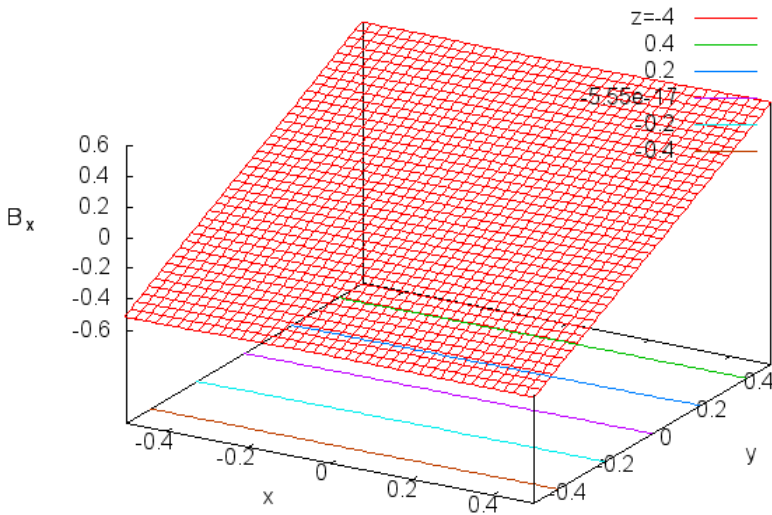
$$B_y = \frac{i}{\sqrt{2}}(B_v - B_u) = i \sum_{j=1}^2 \frac{c_j e_j}{2} [-2ih_j - \ln(1 + e^{\sqrt{2}z - ih_j}) + \ln(1 + e^{\sqrt{2}z + ih_j})]$$

$$B_z = \sqrt{2}B'_z = i \sum_{j=1}^2 \frac{c_j}{\sqrt{2}} [2\ln(2) + 2\sqrt{2}z - \ln(1 + e^{\sqrt{2}z - ih_j}) - \ln(1 + e^{\sqrt{2}z + ih_j})]$$

where  $d_j = \frac{b_j}{\sqrt{2}} + \frac{1}{\sqrt{2}b_j}$ ,  $e_j = \frac{1}{\sqrt{2}b_j} - \frac{b_j}{\sqrt{2}}$  and  $h_j$  is now  $h_j = d_j x + ie_j y$ .

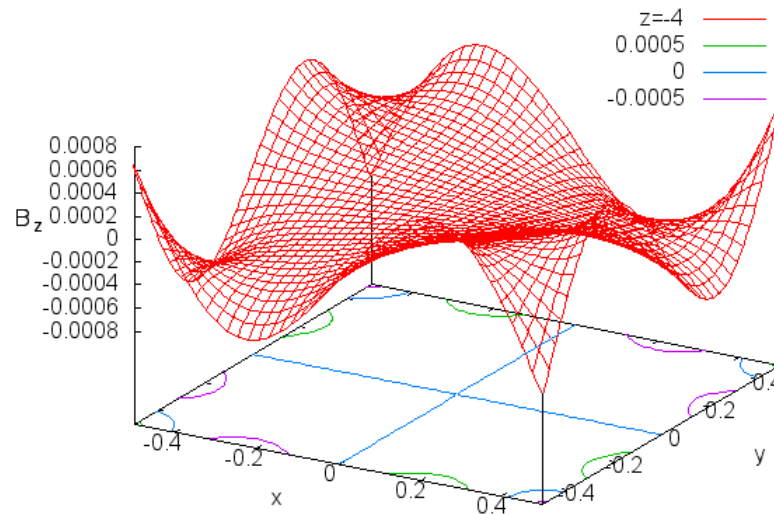
So we are now in a position to look at plots of the various components of the magnetic field and to look at what the meaning of different values of  $b$  is. First we check that the behaviour of the fields is as expected and that the symmetries are what they should be.

# Fringe fields – quadrupoles (7)

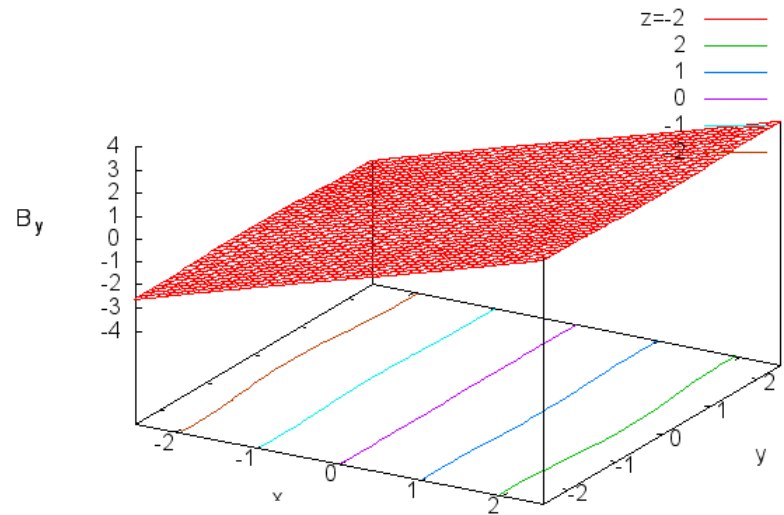
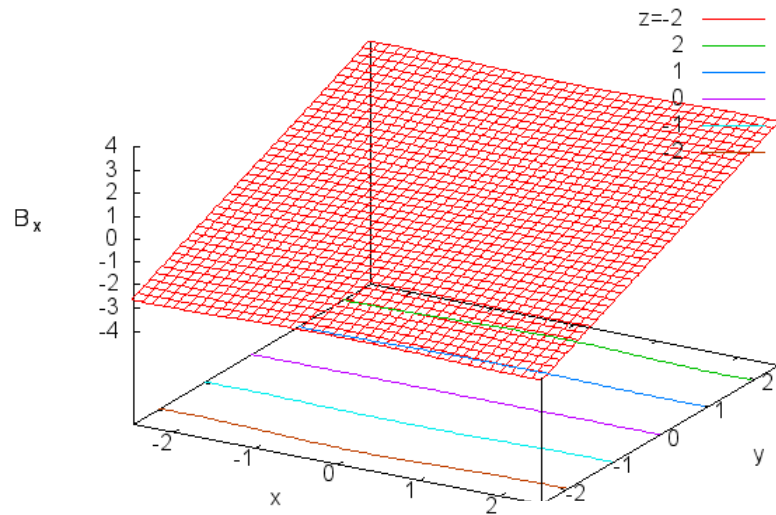


$b = 0.1$  for all movies

$z$  goes from -4 to +4 in arbitrary coordinates

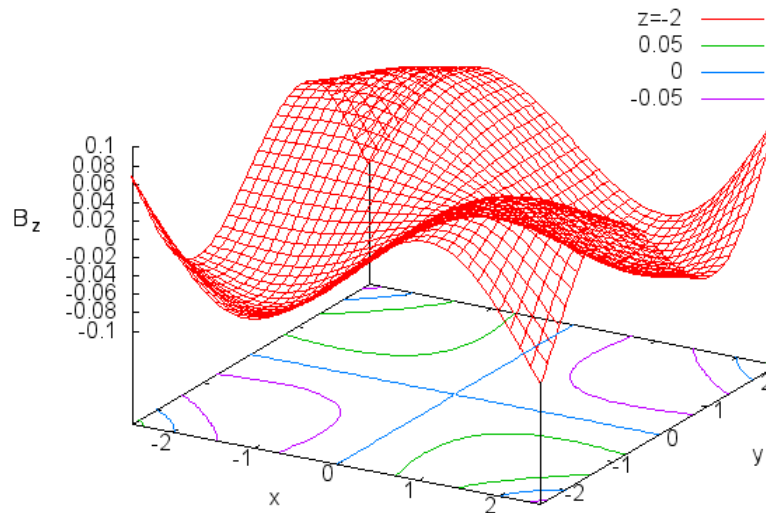


# Fringe fields – quadrupoles (8)



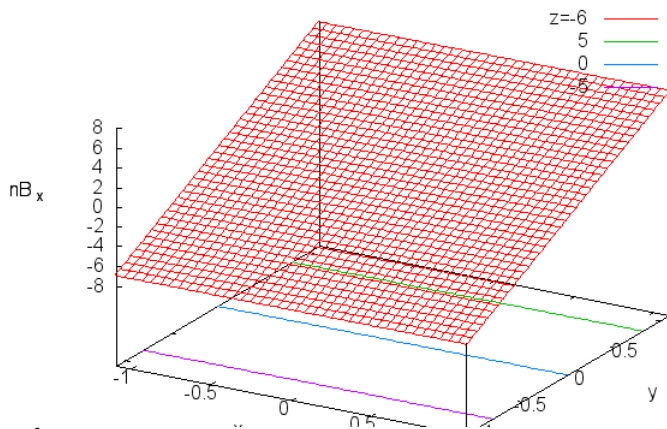
$b = 0.9$  for all movies

$z$  goes from -4 to +4 in arbitrary coordinates



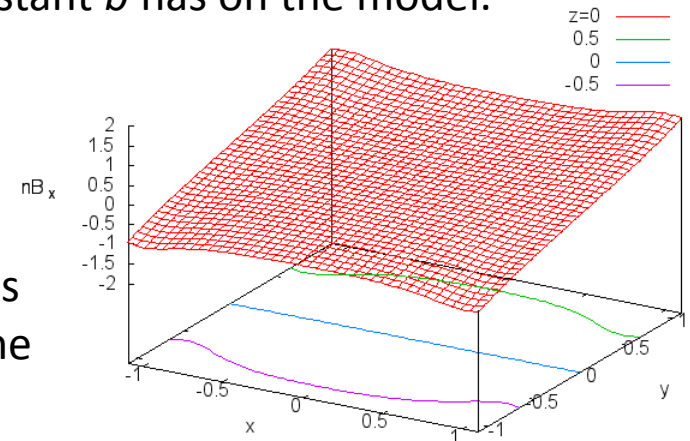
# Fringe fields – quadrupoles (9)

If we treat the singularities or step functions as our limits of validity of the model and identify them with current from the magnet coils, then it is possible to re-scale all the components of the magnetic fields to see the exact effect that the constant  $b$  has on the model.



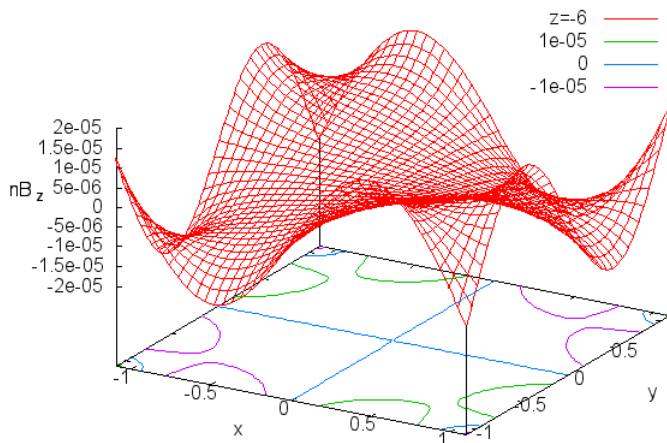
$b = 0.1$

$x$  and  $y$  are now normalised and  $z$  is fixed by how the singularities move as one goes through the fringe field.

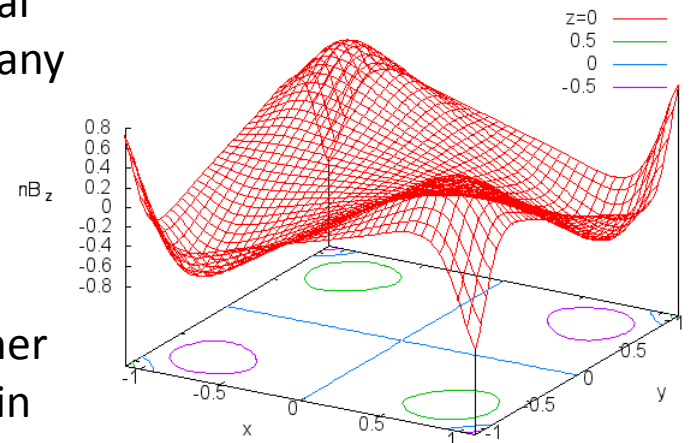


$b = 0.9$

Note that the vertical scale no longer has any meaning.



A higher value of  $b$  means a smoother transition but a higher value of  $B_z$  w.r.t  $B_{x,y}$  in order to achieve it.





# Fringe fields – quadrupoles (10)

So we can write

$$B_x = ic\partial_x\Xi_2(\sqrt{2}z - ih) - ic\partial_x\Xi_2(\sqrt{2}z + ih)$$

$$B_y = ic\partial_y\Xi_2(\sqrt{2}z - ih) - ic\partial_y\Xi_2(\sqrt{2}z + ih)$$

$$B_z = ic\partial_z\Xi_2(\sqrt{2}z - ih) - ic\partial_z\Xi_2(\sqrt{2}z + ih)$$

where  $\Xi_2(x) = \int x - \ln(1 + e^x) + \ln(2)dx = \frac{x^2}{2} + \text{Li}_2(-e^x) + \ln(2)x - \text{Li}_2(-1)$

where  $\text{Li}_2$  is the di-logarithm or poly-logarithm of order 2

And the scalar and vector potential may be written as

$$\varphi = ic[\Xi_2(\sqrt{2}z - ih) - \Xi_2(\sqrt{2}z + ih)]$$

$$A_y = -\frac{\sqrt{2}c}{d}[\Xi_2(\sqrt{2}z - ih) + \Xi_2(\sqrt{2}z + ih)]$$

$$A_z = \frac{ce}{d}[\Xi_2(\sqrt{2}z - ih) - \Xi_2(\sqrt{2}z + ih)]$$

# Fringe fields – quadrupoles (11)

We can fit a function of the form

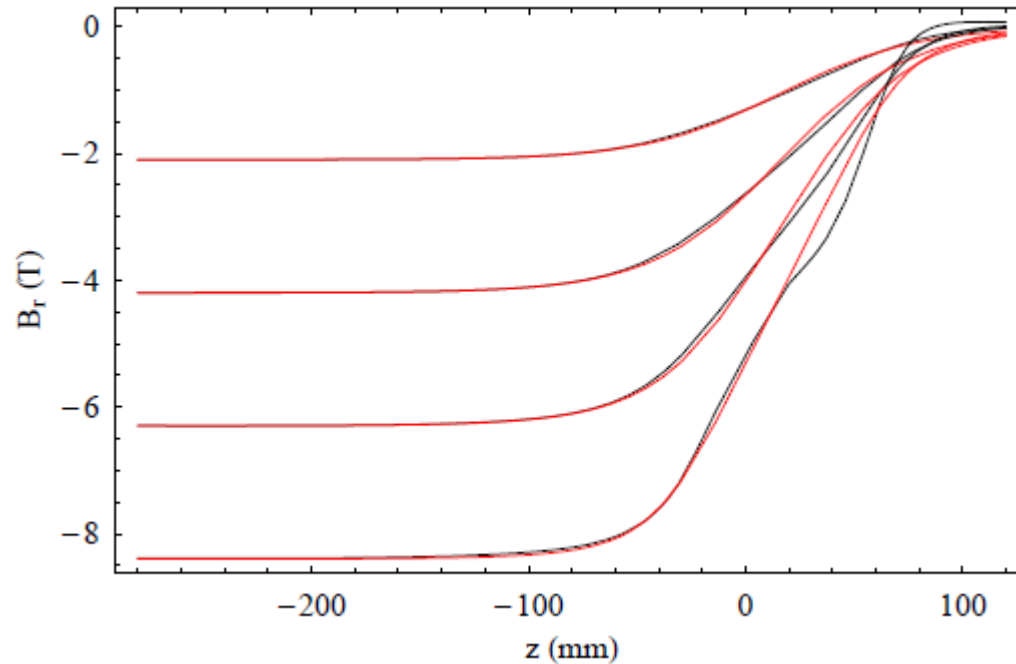
$$g(z) = \frac{a_0}{1 + e^{a_1 + \sqrt{2}a_2 z}}$$

to the fall-off in order to keep things as simple as possible and just illustrate the idea.

*Application to HL-LHC*

| radius of fit | $a_0$    | $a_1$     | $a_2$   |
|---------------|----------|-----------|---------|
| $r_{\max}/10$ | -55.9503 | -0.520120 | 8.98913 |
| $r_{\max}/8$  | -55.9504 | -0.521262 | 9.00549 |
| $r_{\max}/6$  | -55.9505 | -0.523743 | 9.04082 |
| $r_{\max}/4$  | -55.9510 | -0.530875 | 9.14139 |
| $r_{\max}/3$  | -55.9517 | -0.540984 | 9.28116 |
| $r_{\max}/2$  | -55.9539 | -0.570142 | 9.66955 |

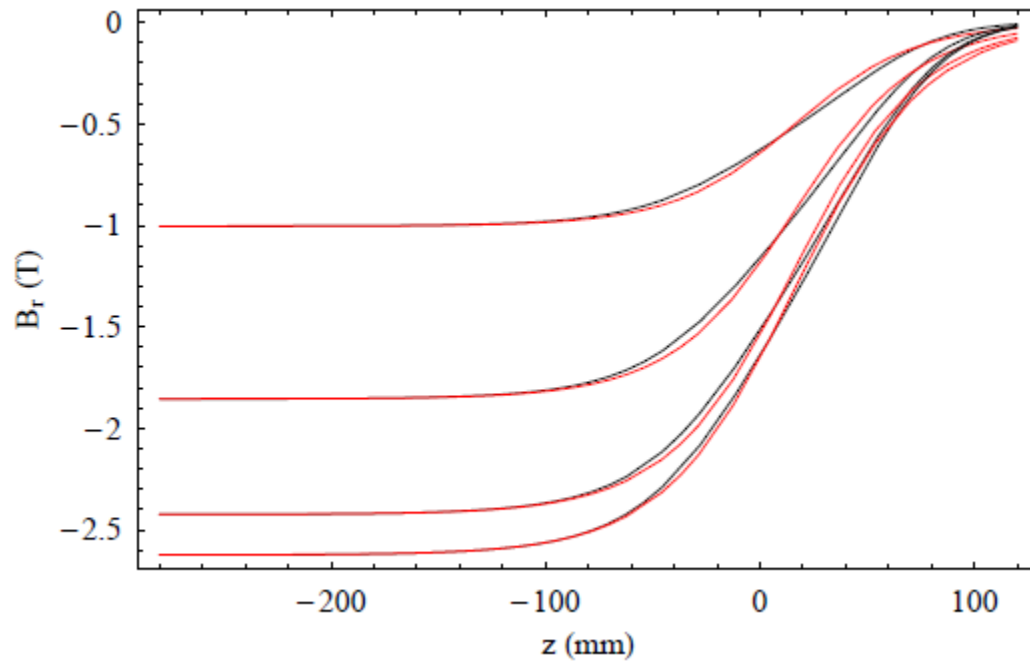
# Fringe fields – quadrupoles (12)



Courtesy: Andy Wolski

Radial component of the magnetic field in an HL-LHC inner triplet quadrupole as a function of position along lines of given radius and with cylindrical polar co-ordinate  $\theta = \pi/4$ . Each pair of black and red lines shows the field at a different radius, from  $r_{\text{max}}/5$  to  $4r_{\text{max}}/5$ , with  $r_{\text{max}} = 75\text{mm}$ . The black lines show the field obtained from the (numerical) magnetic model; the red lines show the analytical model, with fit parameters given in the table with radius of fit  $r_{\text{max}}/10$ , and  $b_1 = 2.5$ . (remember  $b$  is equivalent to  $1/b$ ).

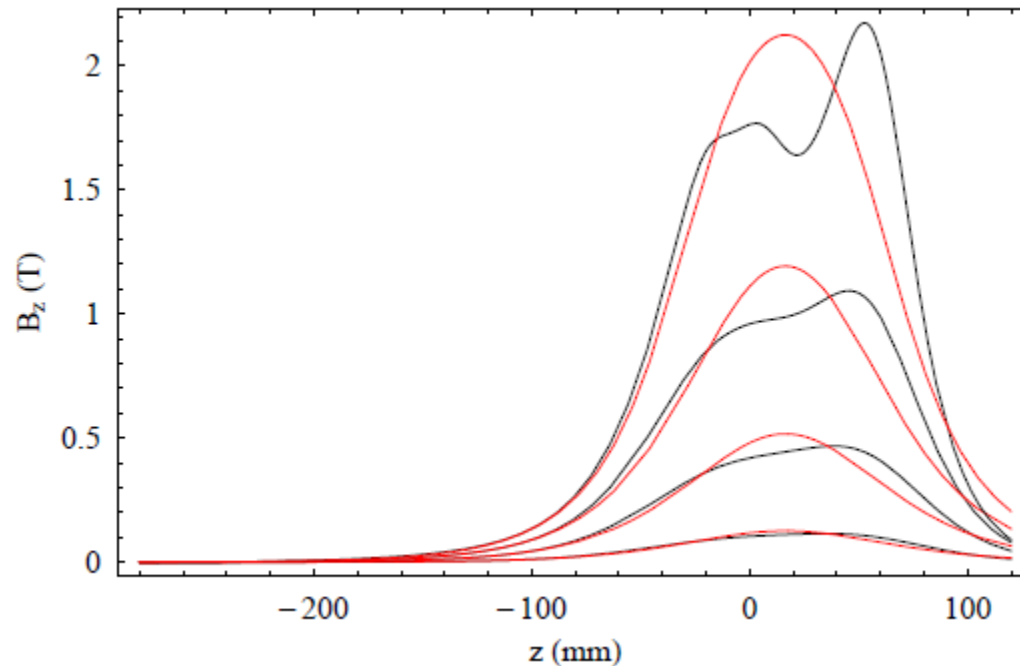
# Fringe fields – quadrupoles (13)



Courtesy: Andy Wolski

Radial component of the magnetic field in an HL-LHC inner triplet quadrupole as a function of position along lines of fixed distance  $r = r_{\text{max}}/4$ . Each pair of black and red lines shows the field at a different value of the cylindrical polar co-ordinate  $\theta$ , from  $\pi/16$  to  $\pi/4$  (in steps of  $\pi/16$ ). The black lines show the field obtained from the (numerical) magnetic model; the red lines show the analytical model, with fit parameters given in the table with radius of fit  $r_{\text{max}}/10$ , and  $b_1 = 2.5$ . (remember  $b$  is equivalent to  $1/b$ ).

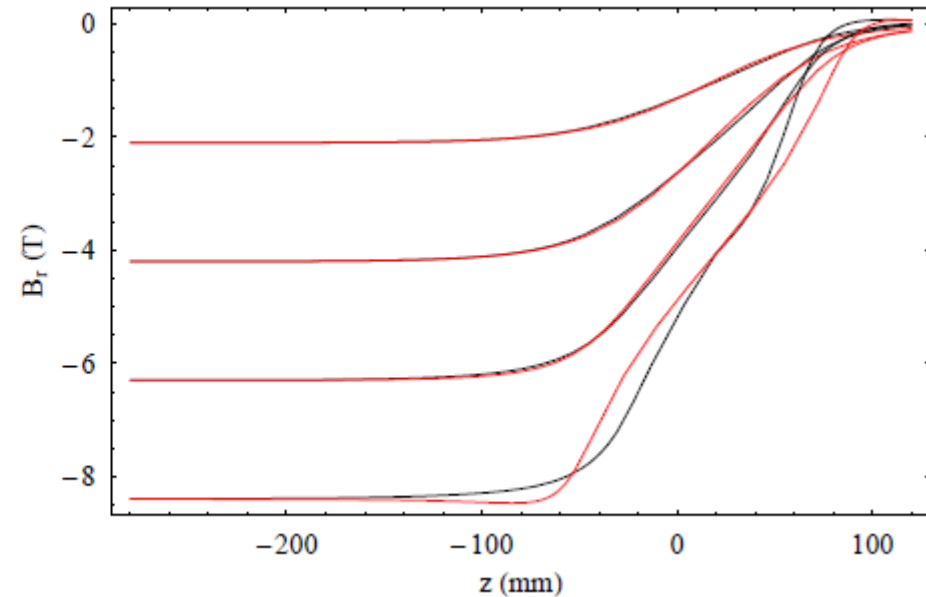
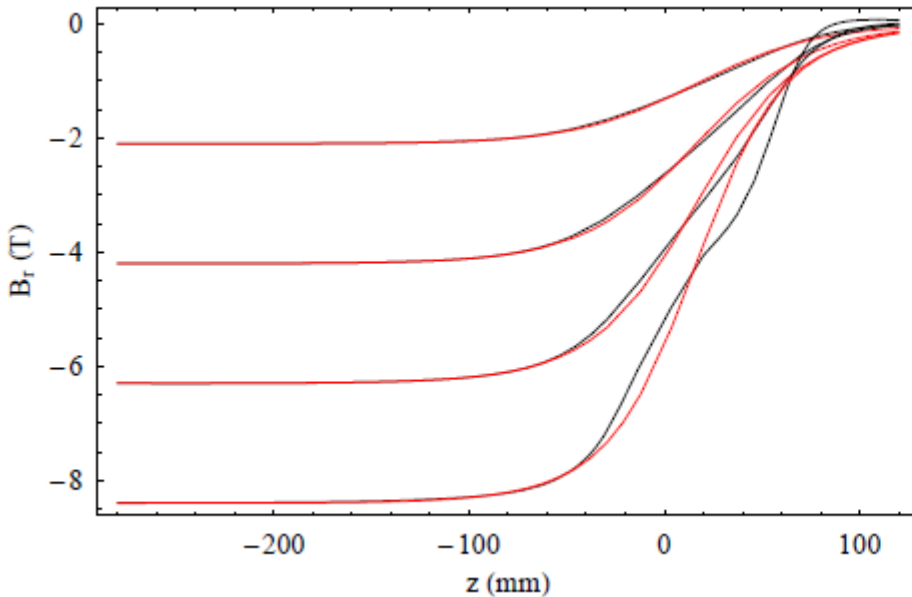
# Fringe fields – quadrupoles (14)



Courtesy: Andy Wolski

Longitudinal component of the magnetic field in an HL-LHC inner triplet quadrupole as a function of position along lines of given radius and with cylindrical polar co-ordinate  $\theta = \pi/4$ . Each pair of black and red lines shows the field at an increasing radius, from  $r_{\text{max}}/5$  to  $4r_{\text{max}}/5$ ,  $r_{\text{max}} = 75\text{mm}$ . The black lines show the field obtained from the (numerical) magnetic model; the red lines show the analytical model, with fit parameters given in the table with radius of fit  $r_{\text{max}}/10$ , and  $b_1 = 2.5$  (remember  $b$  is equivalent to  $1/b$ ).

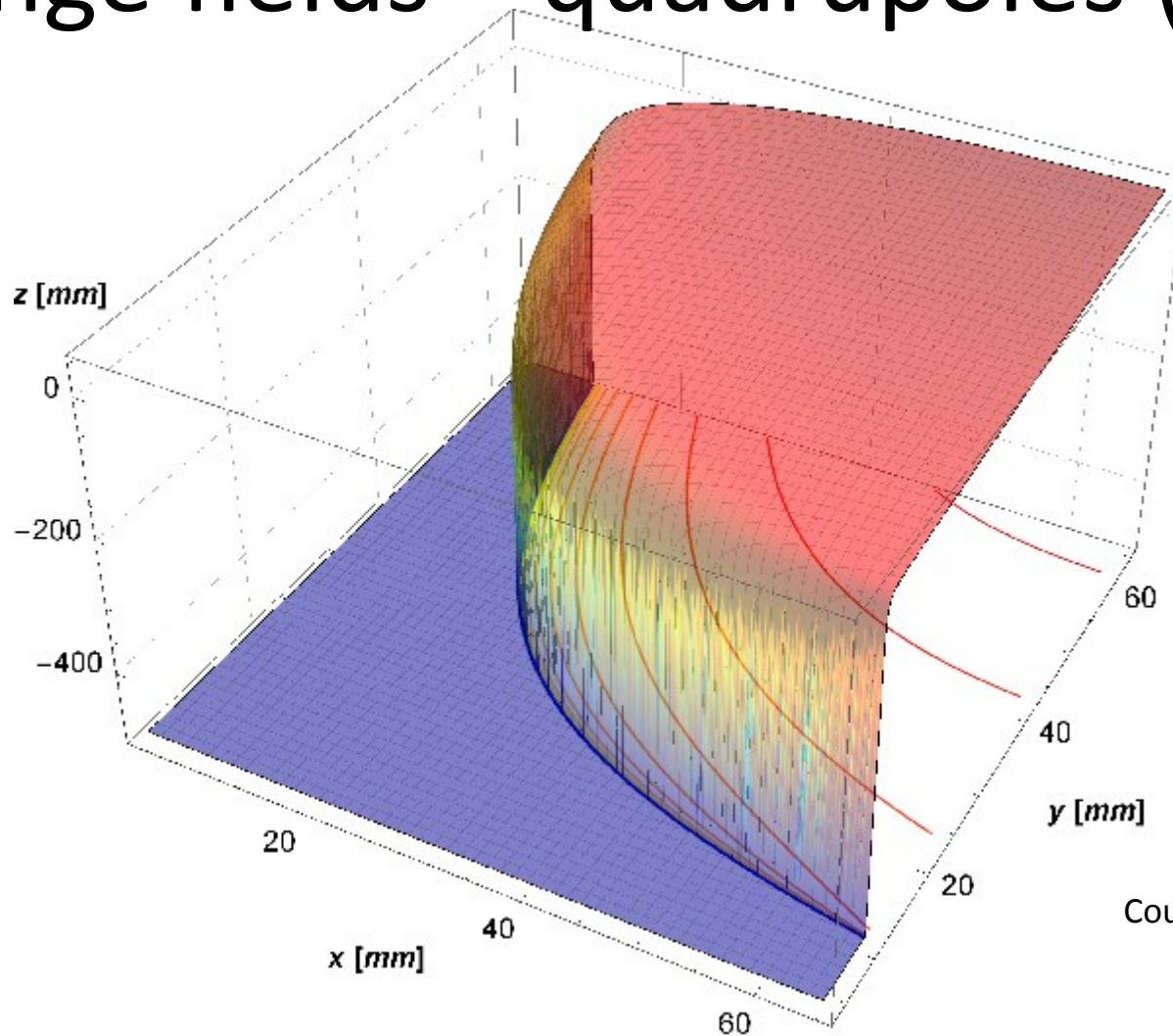
# Fringe fields – quadrupoles (15)



Courtesy: Andy Wolski

Radial component of the magnetic field in an HL-LHC inner triplet quadrupole as a function of position along lines of given radius and with cylindrical polar co-ordinate  $\theta = \pi/4$ . for different values of  $b$  ( $b = 1.5$  left and  $b = 3.5$  right).

# Fringe fields – quadrupoles (16)



Courtesy: Andy Wolski  
and James Jones

Surface of constant scalar potential  $\phi = 0.25 \text{ Tm}$  in a representation of the HL-LHC inner triplet quadrupole, with gradient falling as an Enge function in the fringe field. The parameters of the Enge function correspond to the first line in the table.

# Fringe fields – quadrupoles (17)

We can fit a function of the form

$$g(z) = \frac{a_0}{1 + e^{a_1 + \sqrt{2}a_2 z}}$$

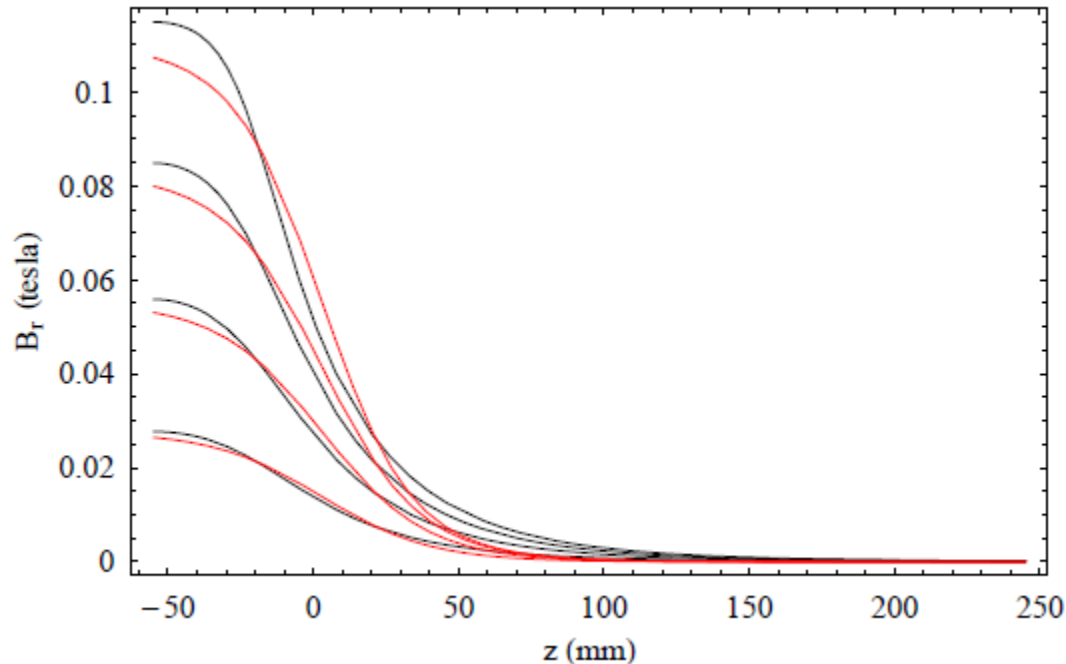
to the fall-off in order to keep things as simple as possible and just illustrate the idea.

*Application to EMMA*

| radius of fit | $a_0$     | $a_1$     | $a_2$   |
|---------------|-----------|-----------|---------|
| $r_{\max}/10$ | 0.0138149 | -0.162670 | 11.2914 |
| $r_{\max}/8$  | 0.0172756 | -0.163023 | 11.3031 |
| $r_{\max}/6$  | 0.0230543 | -0.163794 | 11.3286 |
| $r_{\max}/4$  | 0.0346675 | -0.166075 | 11.4036 |
| $r_{\max}/3$  | 0.0463818 | -0.169480 | 11.5140 |
| $r_{\max}/2$  | 0.0702287 | -0.180796 | 11.8694 |



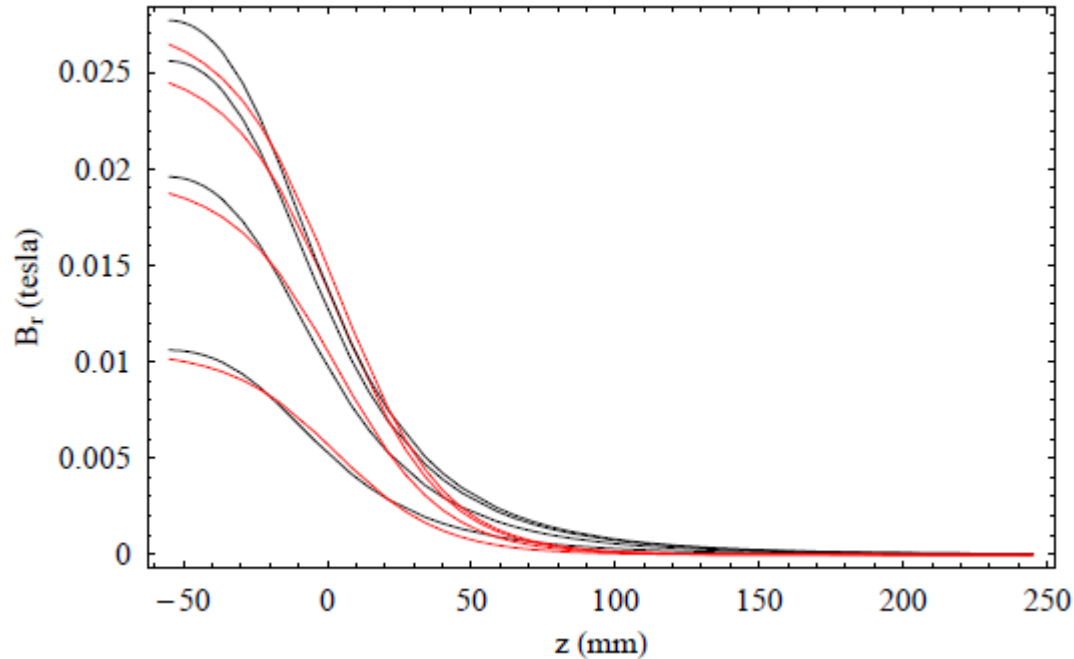
# Fringe fields – quadrupoles (18)



Courtesy: Andy Wolski

Radial component of the magnetic field in an EMMA F quadrupole as a function of position along lines of given radius and with cylindrical polar co-ordinate  $\theta = \pi/4$ . Each pair of black and red lines shows the field at a different radius, from  $r_{\max}/5$  to  $4r_{\max}/5$ , with  $r_{\max} = 36\text{mm}$ . The black lines show the field obtained from the (numerical) magnetic model; the red lines show the analytical model, with fit parameters given in the table with radius of fit  $r_{\max}/10$ , and  $b_1 = 1.8$  (remember  $b$  is equivalent to  $1/b$ ). The centre of the quadrupole is at the far left of the plot,  $z = -53.6\text{mm}$ .

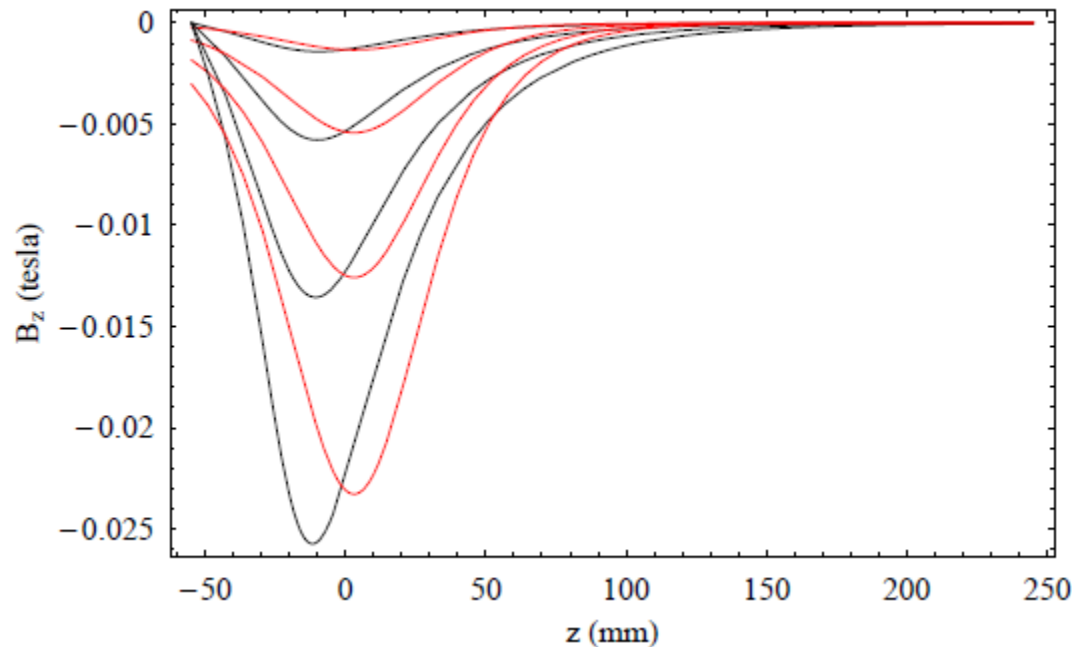
# Fringe fields – quadrupoles (19)



Courtesy: Andy Wolski

Radial component of the magnetic field in an EMMA F quadrupole as a function of position along lines of fixed distance  $r = r_{\text{max}}/4$ . Each pair of black and red lines shows the field at a different value of the cylindrical polar co-ordinate  $\theta$ , from  $\pi/16$  to  $\pi/4$  (in steps of  $\pi/16$ ). The black lines show the field obtained from the (numerical) magnetic model; the red lines show the analytical model, with fit parameters given in Table 2 with radius of fit  $r_{\text{max}}/10$ , and  $b_1 = 1.8$  (remember  $b$  is equivalent to  $1/b$ ). The centre of the quadrupole is at the far left of the plot,  $z = -53.6\text{mm}$ .

# Fringe fields – quadrupoles (20)



Courtesy: Andy Wolski

Longitudinal component of the magnetic field in an EMMA F quadrupole as a function of position along lines of given radius and with cylindrical polar co-ordinate  $\theta = \pi/4$ . Each pair of black and red lines shows the field at an increasing radius, from  $r_{\text{max}}/5$  to  $4r_{\text{max}}/5$ , with  $r_{\text{max}} = 36\text{mm}$ . The black lines show the field obtained from the (numerical) magnetic model; the red lines show the analytical model, with fit parameters given in the table with radius of fit  $r_{\text{max}}/10$ , and  $b_1 = 1.8$  (remember  $b$  is equivalent to  $1/b$ ). The centre of the quadrupole is at the far left of the plot,  $z = -53.6\text{mm}$ .

# Fringe fields – multipoles (1)

According to the multipole geometry, we need  $n = n$  in the defining fringe field equations. We use the same functions  $F$  and  $G$  as for the previous cases. Therefore, the equations we have to satisfy for the components of the magnetic field (for  $B_u$ ,  $B_v$  and  $B_z$  in order) are

*Multipole equations*

$$-ic_1 b_1 \left( \frac{u}{b_1} + b_1 v + iz \right)^n - ic_1 b_1 \left( \frac{u}{b_1} + b_1 v - iz \right)^n - ic_2 b_2 \left( \frac{u}{b_2} + b_2 v + iz \right)^n - ic_2 b_2 \left( \frac{u}{b_2} + b_2 v - iz \right)^n \\ + \dots - ic_{n+1} b_{n+1} \left( \frac{u}{b_{n+1}} + b_{n+1} v + iz \right)^n - ic_n b_{n+1} \left( \frac{u}{b_{n+1}} + b_{n+1} v - iz \right)^n = iv^n$$

$$-i \frac{c_1}{b_1} \left( \frac{u}{b_1} + b_1 v + iz \right)^n - i \frac{c_1}{b_1} \left( \frac{u}{b_1} + b_1 v - iz \right)^n - i \frac{c_2}{b_2} \left( \frac{u}{b_2} + b_2 v + iz \right)^n - i \frac{c_2}{b_2} \left( \frac{u}{b_2} + b_2 v - iz \right)^n \\ + \dots - i \frac{c_{n+1}}{b_{n+1}} \left( \frac{u}{b_{n+1}} + b_{n+1} v + iz \right)^n - i \frac{c_{n+1}}{b_{n+1}} \left( \frac{u}{b_{n+1}} + b_{n+1} v - iz \right)^n = -iu^n$$

$$-ic_1 \left( \frac{u}{b_1} + b_1 v + iz \right)^n - ic_1 \left( \frac{u}{b_1} + b_1 v - iz \right)^n - ic_2 \left( \frac{u}{b_2} + b_2 v + iz \right)^n - ic_2 \left( \frac{u}{b_2} + b_2 v - iz \right)^n \\ + \dots - ic_{n+1} \left( \frac{u}{b_{n+1}} + b_{n+1} v + iz \right)^n - ic_{n+1} \left( \frac{u}{b_{n+1}} + b_{n+1} v - iz \right)^n = 0.$$

Giving the following  $n+2$  equations to satisfy

# Fringe fields – multipoles (2)

$$c_1 b_1^\alpha + c_2 b_2^\alpha + \dots + c_{n+1} b_{n+1}^\alpha = 0$$

$$\frac{c_1}{b_1^\alpha} + \frac{c_2}{b_2^\alpha} + \dots + \frac{c_{n+1}}{b_{n+1}^\alpha} = 0$$

$$2c_1 b_1^n + 2c_2 b_2^n + \dots + 2c_{n+1} b_{n+1}^n = -1$$

$$2\frac{c_1}{b_1^n} + 2\frac{c_2}{b_2^n} + \dots + 2\frac{c_{n+1}}{b_{n+1}^n} = 1$$

with  $\alpha = n - 2m$

Which can be solved in an analogous way to the octupole case (only more complicated). The solution now involves  $n-2$  independent constants

$$c_1 = -\frac{b_1^{n-2}}{2 \prod_{j=2}^{n+1} (b_1^2 - b_j^2)}, \quad c_2 = -\frac{b_2^{n-2}}{2 \prod_{j=1, j \neq 2}^{n+1} (b_2^2 - b_j^2)}, \quad \dots,$$

$$c_n = -\frac{b_n^{n-2}}{2 \prod_{j=1}^{n-1} (b_n^2 - b_j^2)}, \quad b_1 = \pm \frac{i^{n-2}}{\prod_{j=2}^{n+1} b_j}.$$

# Fringe fields – multipoles (3)

*Potentials for the multipole solution*

$$B_x = -icd[(i)^n \Xi_n(\sqrt{2}z - ih) + (-i)^n \Xi_n(\sqrt{2}z + ih)]$$

$$B_y = ce[(i)^n \Xi_n(\sqrt{2}z - ih) + (-i)^n \Xi_n(\sqrt{2}z + ih)]$$

$$B_z = \sqrt{2}c[(i)^n \Xi_n(\sqrt{2}z - ih) - (-i)^n \Xi_n(\sqrt{2}z + ih)]$$

with

$$\Xi_n(x) = \frac{x^n}{n!} + \text{Li}_n(-e^x) - \sum_{j=1}^n \text{Li}_j(-1) \frac{x^{n-j}}{(n-j)!}$$

where  $\text{Li}_n$  is the di-logarithm or poly-logarithm of order  $n$

giving the potentials

$$\varphi = c[(i)^n \Xi_{n+1}(\sqrt{2}z - ih) - (-i)^n \Xi_{n+1}(\sqrt{2}z + ih)]$$

$$A_y = -\frac{\sqrt{2}c}{d}[(i)^n \Xi_{n+1}(\sqrt{2}z - ih) + (-i)^n \Xi_{n+1}(\sqrt{2}z + ih)]$$

$$A_z = \frac{ce}{d}[(i)^n \Xi_{n+1}(\sqrt{2}z - ih) + (-i)^n \Xi_{n+1}(\sqrt{2}z + ih)]$$

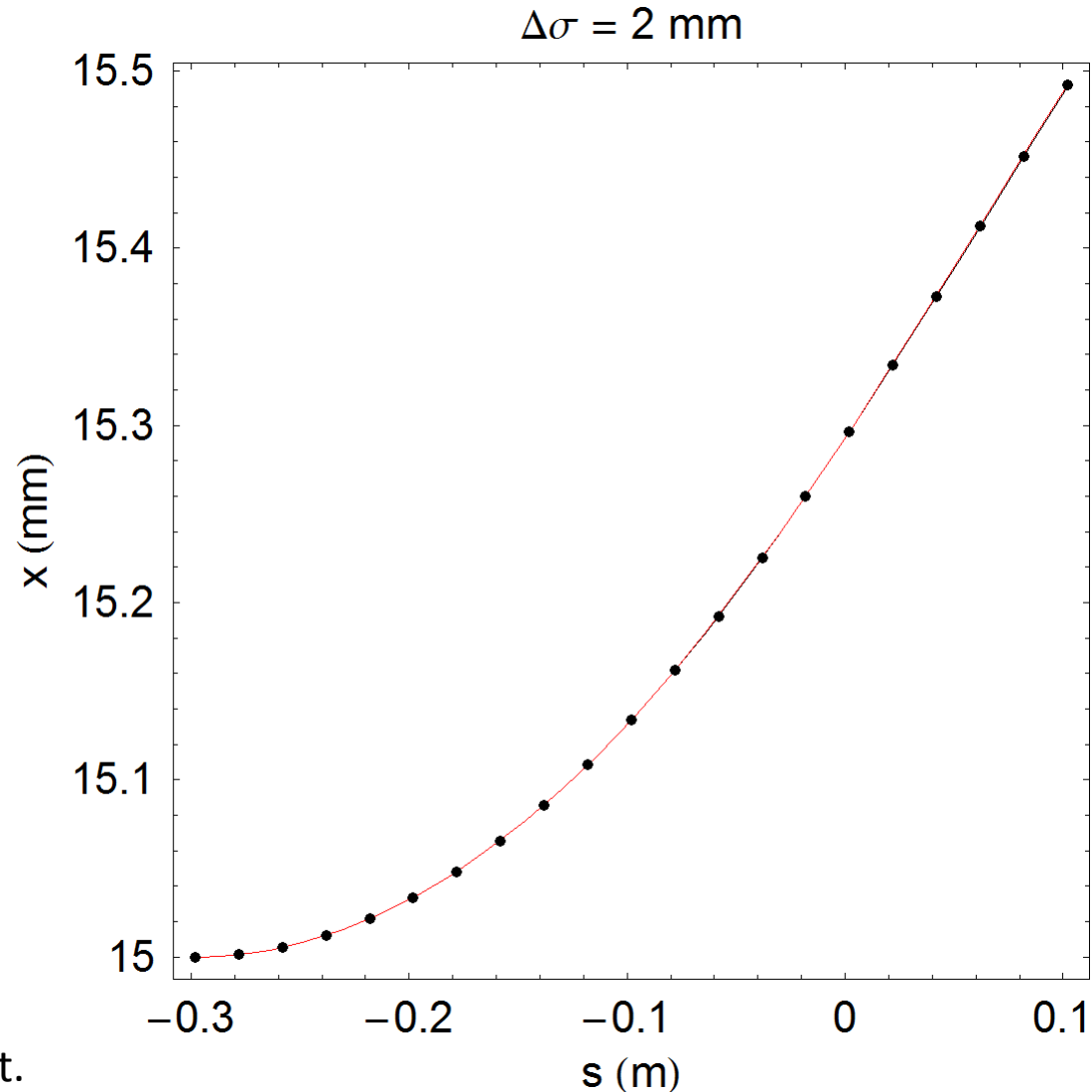
# Fringe fields – trajectories (1)

HL-LHC Inner triplet

Red line: trajectory obtained by a Runge-Kutta integration of the Newtonian equations of motion ( $dp/dt = q \mathbf{v} \times \mathbf{B}$ ), using the field map provided by CERN (i.e. no fitting).

Black line: trajectory obtained by a Runge-Kutta integration of the Hamilton equations of motion, using the vector potential obtained by fitting the field map with an Enge function roll-off of the gradient.

Black points: trajectory obtained by a Wu-Forest-Robin (explicit symplectic integration) of the Hamilton equations of motion, using the vector potential obtained by fitting the field map with an Enge function roll-off of the gradient.



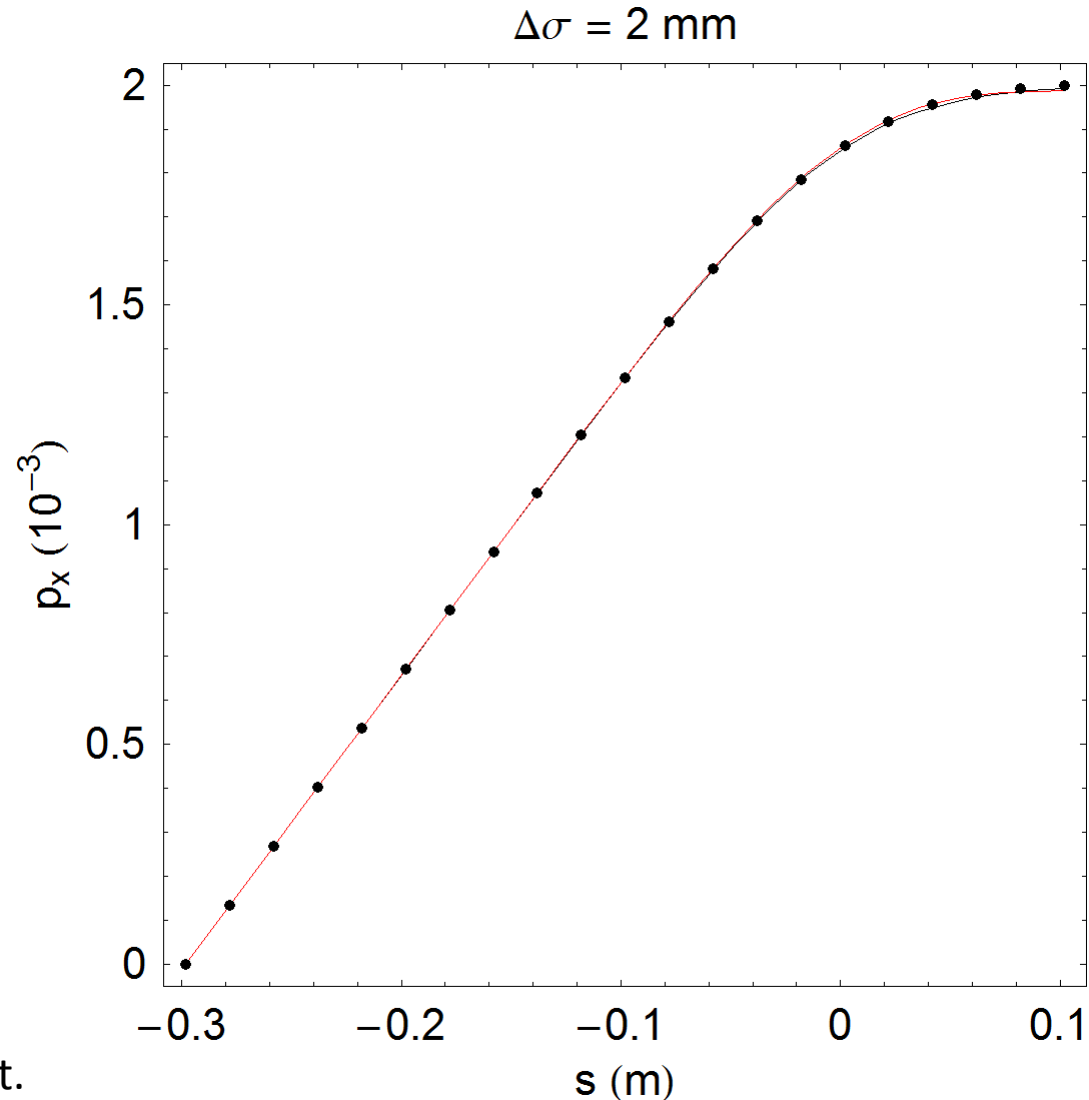
# Fringe fields – trajectories (2)

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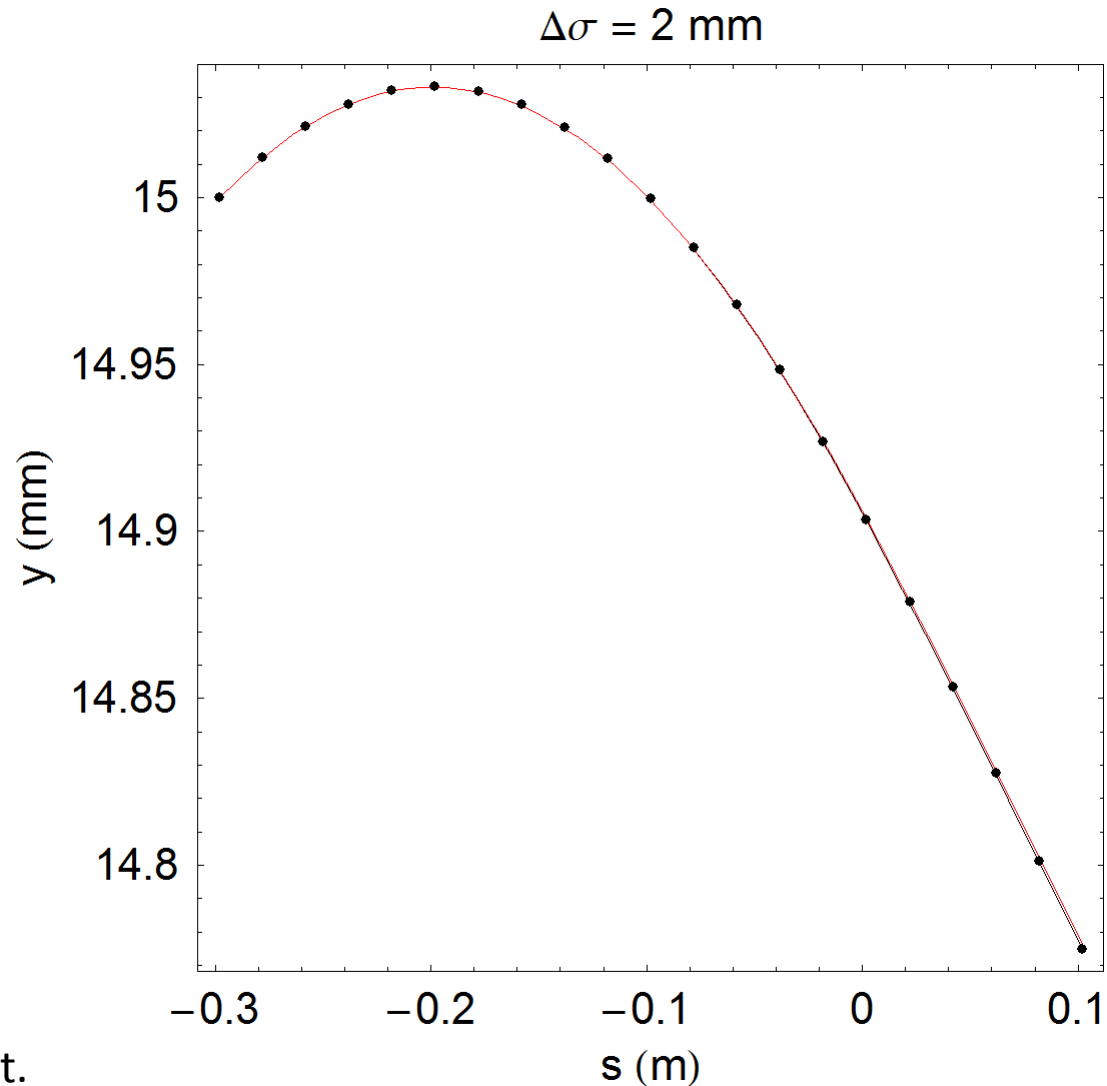
# Fringe fields – trajectories (3)

HL-LHC Inner triplet

Red line: trajectory obtained by a Runge-Kutta integration of the Newtonian equations of motion ( $dp/dt = q \mathbf{v} \times \mathbf{B}$ ), using the field map provided by CERN (i.e. no fitting).

Black line: trajectory obtained by a Runge-Kutta integration of the Hamilton equations of motion, using the vector potential obtained by fitting the field map with an Enge function roll-off of the gradient.

Black points: trajectory obtained by a Wu-Forest-Robin (explicit symplectic integration) of the Hamilton equations of motion, using the vector potential obtained by fitting the field map with an Enge function roll-off of the gradient.



# Fringe fields – trajectories (4)

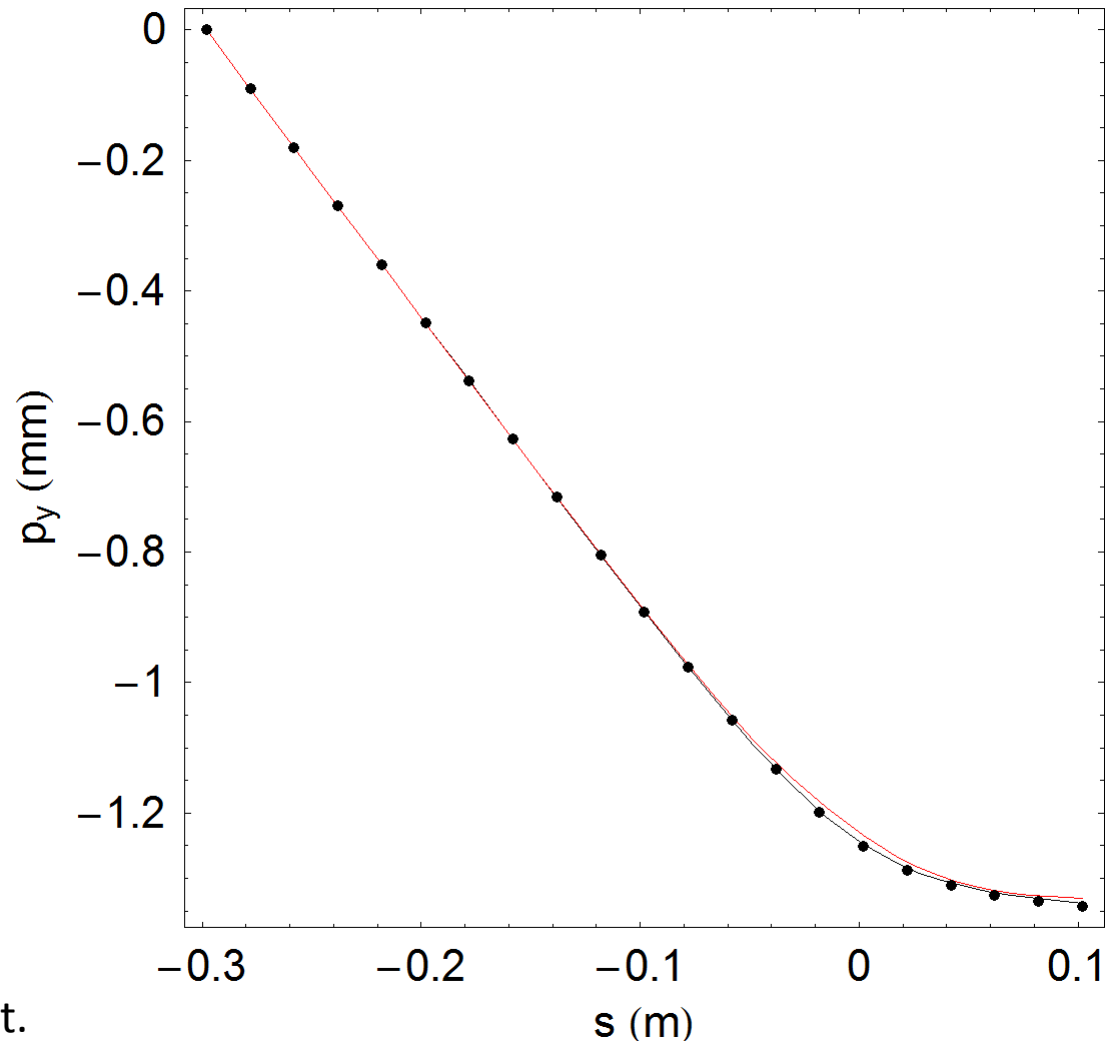
HL-LHC Inner triplet

$\Delta\sigma = 2 \text{ mm}$

Red line: trajectory obtained by a Runge-Kutta integration of the Newtonian equations of motion ( $dp/dt = q \mathbf{v} \times \mathbf{B}$ ), using the field map provided by CERN (i.e. no fitting).

Black line: trajectory obtained by a Runge-Kutta integration of the Hamilton equations of motion, using the vector potential obtained by fitting the field map with an Enge function roll-off of the gradient.

Black points: trajectory obtained by a Wu-Forest-Robin (explicit symplectic integration) of the Hamilton equations of motion, using the vector potential obtained by fitting the field map with an Enge function roll-off of the gradient.



# Simple Expression (Quadrupole)

It is possible to take the limit as  $b \rightarrow 1$  to obtain simpler expressions which still satisfy Maxwell but are much simpler to implement in tracking codes. This is currently being implemented in the General Particle Tracer (GPT) code by the authors Bas and Marieke Van Der Geer. If this limit is taken, then the additional degree of freedom which dictates the rapidity of the fringe field fall off as one goes off-axis is no longer available. However, this does imply a lack of generality but it is not an important one in most cases, though in both cases considered in this talk this constant is indeed required.

$$\begin{aligned} & \frac{1}{4} \left( -3y - 2 \operatorname{ArcTan}[1 + e^z \cos[y], -e^z \sin[y]] + \frac{y \sinh[z]}{\cos[x] + \cosh[z]} \right) \\ & \frac{1}{4} \left( -3x - 2 \operatorname{ArcTan}[1 + e^z \cos[x], -e^z \sin[x]] + \frac{x \sinh[z]}{\cos[y] + \cosh[z]} \right) \\ & \frac{1}{4} \left( \frac{y \sin[x]}{\cos[x] + \cosh[z]} + \frac{x \sin[y]}{\cos[y] + \cosh[z]} \right) \end{aligned}$$

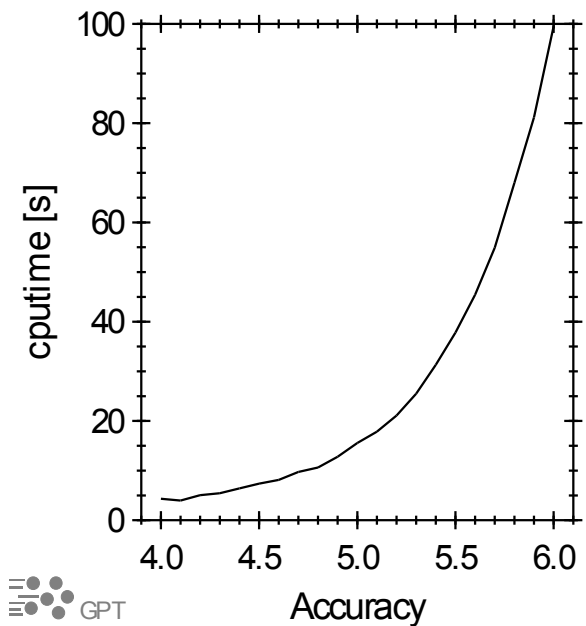
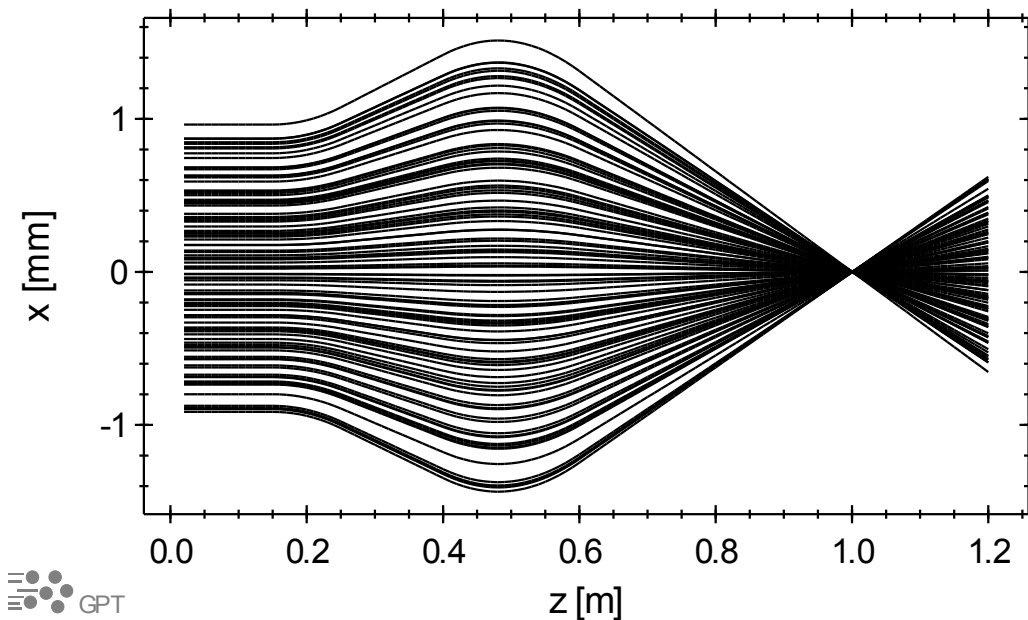
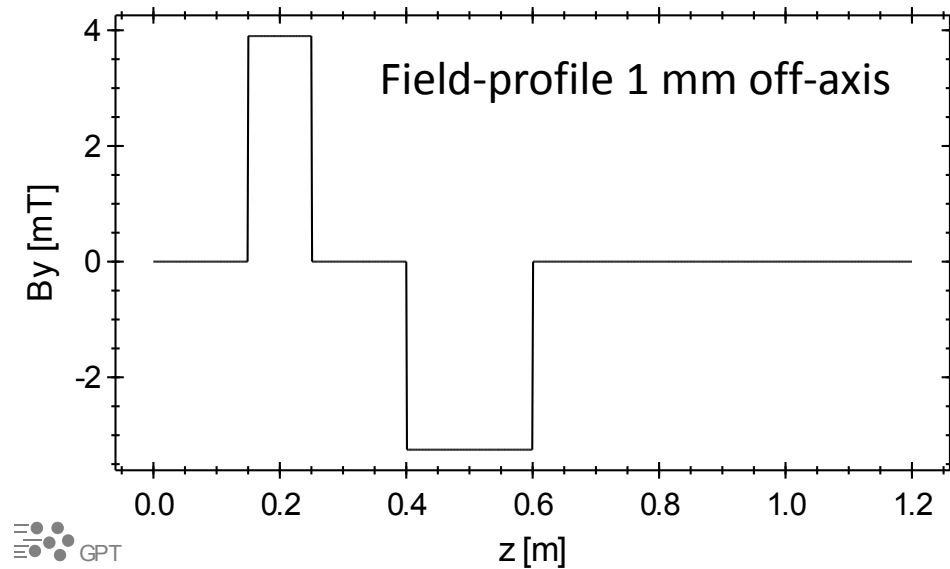
Courtesy: Bas Van Der Geer

# GPT Application (1)

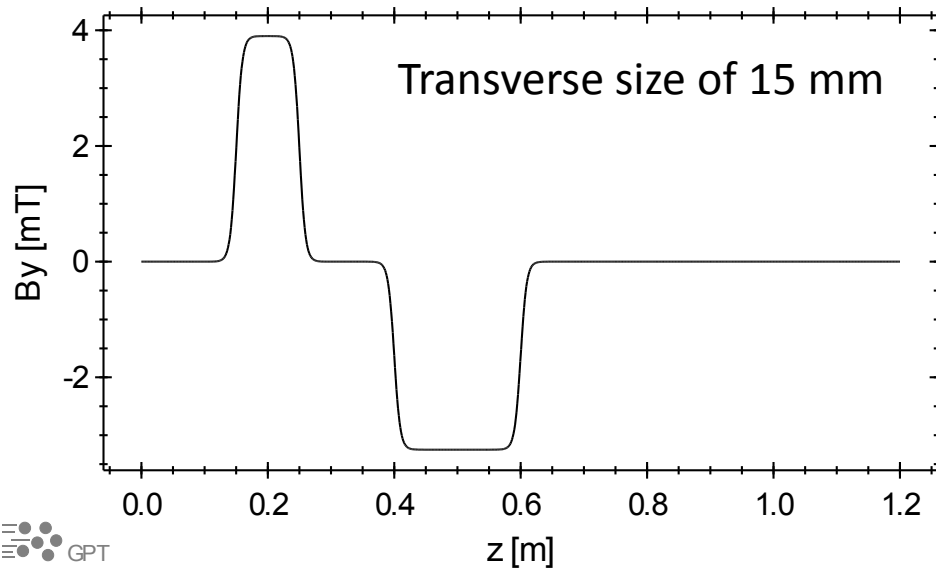
Equations: Bruno Muratori  
Implementation: Pulsar Physics

Test case:

- Two quadrupoles (GPT tutorial)
- 10.000 particles
- 10 mm bunch length

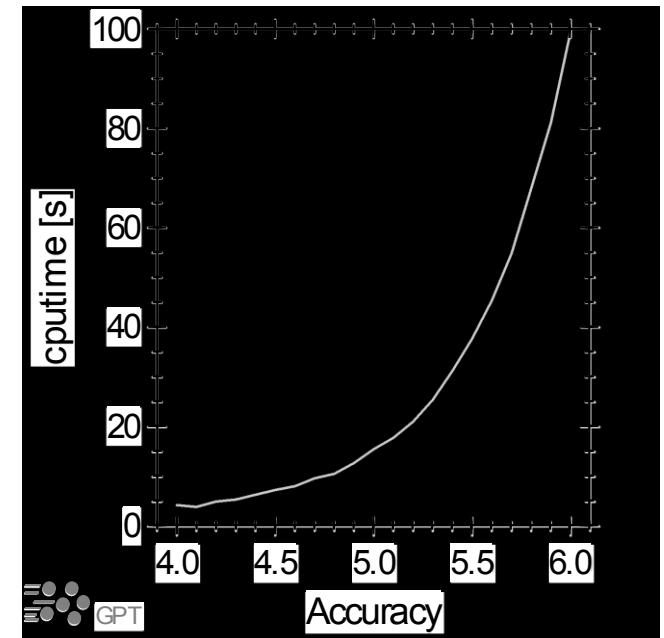
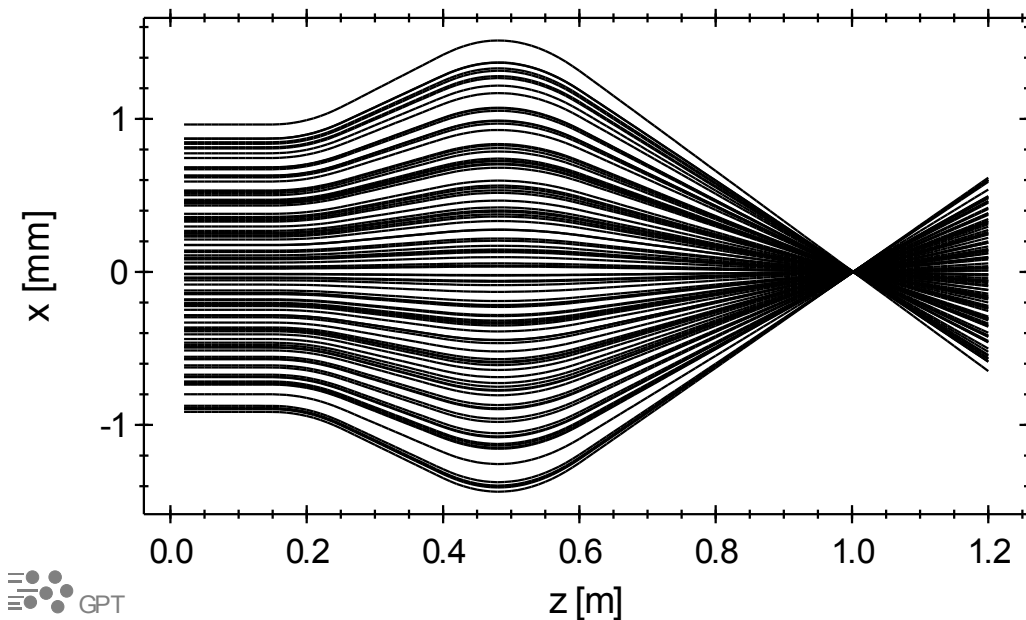


# GPT Application (2)



Conclusion:

- It works
- Gives **same** results
- But much **faster**
- Difference depends on accuracy



# CFM – preliminaries (1)

Cartesian coordinates (X, Y, Z) to cylindrical (x, y, s):

$$X = (x + \rho) \cos\left(\frac{s}{\rho}\right) - \rho, \quad Y = y, \quad \text{and} \quad Z = (x + \rho) \sin\left(\frac{s}{\rho}\right)$$

For a normal coordinate transformation, the components of field also transform:

$$B_x = B_Z \sin\left(\frac{s}{\rho}\right) + B_X \cos\left(\frac{s}{\rho}\right), \quad B_y = B_Y, \quad \text{and} \quad B_s = B_Z \cos\left(\frac{s}{\rho}\right) - B_X \sin\left(\frac{s}{\rho}\right)$$

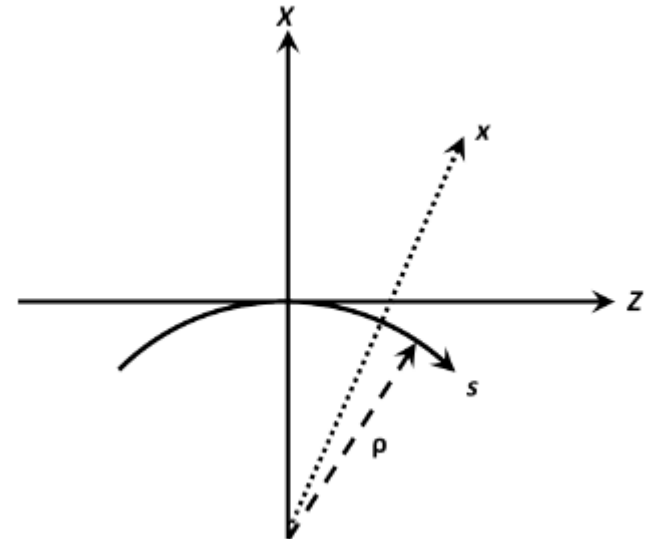
The magneto-static Maxwell equations become:

$$\frac{hB_x}{(1+hx)} + \partial_x B_x + \partial_y B_y + \frac{1}{(1+hx)} \partial_s B_s = 0,$$

$$\partial_y B_s - \frac{1}{(1+hx)} \partial_s B_y = 0,$$

$$\frac{1}{(1+hx)} \partial_s B_x - \partial_x B_s - \frac{hB_s}{(1+hx)} = 0,$$

where  $h = 1/\rho$



# CFM – preliminaries (2)

The components of the vector field also transform according to

$$A_x = A_Z \sin\left(\frac{s}{\rho}\right) + A_X \cos\left(\frac{s}{\rho}\right), \quad A_y = A_Y, \quad \text{and} \quad A_s = A_Z \cos\left(\frac{s}{\rho}\right) - A_X \sin\left(\frac{s}{\rho}\right)$$

The Maxwell equations to be solved in these coordinates are

$$B_x = \partial_y A_s - \frac{1}{(1+hx)} \partial_s A_y,$$

$$B_y = \frac{1}{(1+hx)} \partial_s A_x - \partial_x A_s - \frac{hA_s}{(1+hx)},$$

$$B_s = \partial_x A_y - \partial_y A_x.$$

We further choose the following simplification

$$A_x = A_y = 0$$

which we can do for the time being. Looking at the simplest case first:

## ***Dipoles:***

The equations to be solved simplify to  $\partial_y A_s = 0$ , and  $-\partial_x A_s - \frac{hA_s}{(1+hx)} = B_0$ ,

with the requirements that  $B_x = B_s = 0$  and  $B_y = B_0$  where  $B_0$  is a constant

# CFM – preliminaries (3)

The solution is quite simple and can be written as  $A_s = -B_0x + \frac{B_0hx^2}{2(1+hx)}$ ,

If we further note that  $\frac{k}{(1+hx)}$ , corresponds to a gauge term and may be removed then

the solution becomes  $A_s = -\frac{B_0}{2h}(1+hx)$ ,

In general  $\vec{B} = \vec{\nabla} \times \vec{A}$  and  $\vec{B} = \vec{\nabla} \varphi$  give (with  $A_x = A_y = 0$ ):

1

$$(\partial_x^2 + \partial_y^2)A_s + \frac{h}{(1+hx)}\partial_x A_s - \frac{h^2}{(1+hx)^2}A_s = 0. \quad \text{for } A_s \text{ and}$$

$$\vec{\nabla} \varphi = \partial_x \varphi \vec{e}_x + \partial_y \varphi \vec{e}_y + \frac{1}{(1+hx)}\partial_s \varphi \vec{e}_s, \quad \text{so}$$

$$(\partial_x^2 + \partial_y^2)\varphi + \frac{1}{(1+hx)^2}\partial_s^2 \varphi + \frac{h}{(1+hx)}\partial_x \varphi = 0.$$

If we further ask that nothing depends on 's', which makes sense outside solenoids, then:

2

$$(\partial_x^2 + \partial_y^2)\varphi + \frac{h}{(1+hx)}\partial_x \varphi = 0.$$



# CFM – preliminaries (4)

We can also try to solve the Maxwell equations directly If we assume

$$B_x = f(x, y), \quad B_y = g(x, y),$$

then

$$(\partial_x^2 + \partial_y^2)f + \frac{h}{(1+hx)}\partial_x f - \frac{h^2}{(1+hx)^2}f = 0, \quad \text{and} \quad (\partial_x^2 + \partial_y^2)g + \frac{h}{(1+hx)}\partial_x g = 0,$$

Assume following form of  $f(x, y)$  for simplicity  $f(x, y) = f_1(x)f_2(y)$ , so

1,2

$$\partial_x^2 f_1 + \frac{h}{(1+hx)}\partial_x f_1 - \frac{h^2}{(1+hx)^2}f_1 = -kf_1,$$

Periodic solutions do not really make sense so we are left with:

$$\partial_y^2 f_2 = kf_2,$$

$$f_2 = c_1 y + c_2,$$

$c_1$  &  $c_2$  const.

The first equation can be reduced to:

$$\partial_x \left( \partial_x f_1 + \frac{h}{(1+hx)}f_1 \right) = 0, \quad \rightarrow \quad \partial_x f_1 + \frac{h}{(1+hx)}f_1 = d_1,$$

$d_1$  &  $d_2$  const.

$$\rightarrow \partial_x [(1+hx)f_1] = d_1(1+hx),$$

so

$$f_1 = \frac{d_1 x + d_1 h \frac{x^2}{2} + d_2}{(1+hx)},$$

# CFM – preliminaries (5)

We now need to substitute these expressions back into the Maxwell equations to further constrain them

$$\partial_x g = c_1 f_1 = \frac{c_1(d_1 x + d_1 h \frac{x^2}{2} + d_2)}{(1 + hx)},$$

from which we have:

$$g = \frac{c_1 d_1 x^2}{4} + \frac{c_1 d_1 x}{2h} + \frac{1}{h} \left( c_1 d_2 - \frac{c_1 d_1}{2h^2} \right) \ln(1 + hx) + p(y).$$

Substituting into the remaining equation and gathering terms in  $x$  and  $y$ , we have:

$$\partial_x f_1 + \frac{h}{(1 + hx)} f_1 = e_1, \quad \text{and} \quad \frac{\partial_y g}{f_2} = -e_1 = \frac{\partial_y p}{f_2}.$$

So we can write:

$$\partial_y p = -d_1 f_2 = -d_1(c_1 y + c_2) \quad \rightarrow \quad p = -d_1 c_1 \frac{y^2}{2} - d_1 c_2 y + c_3,$$

# CFM – dipole & quadrupole (1)

We need to satisfy

$$B_x = B_1 y + \dots, \quad B_y = B_0 + B_1 x + \dots, \quad B_s = 0.$$

with the function  $f(x,y)$  and  $g(x,y)$  therefore, we require the following expansion:

$$\frac{1}{(1+hx)} = 1 - hx + (hx)^2 - (hx)^3 + \dots + (-1)^n (hx)^n,$$

So we can write  $B_x$  as

$$B_x = d_2(c_2 + c_1 y) + (d_1 - d_2 h)(c_2 + c_1 y)x + \sum_{n=1}^{\infty} (-1)^n \left[ \frac{d_1 h^n}{2} - d_2 h^{n+1} \right] (c_2 + c_1 y) x^{n+1}.$$

From which we easily find that

$$c_2 = d_1 = 0$$

Substituting this into the equation for  $g(x,y)$ , we have:

# CFM – dipole & quadrupole (2)

$$g = \frac{c_1 d_2}{h} \ln(1 + hx) + c_3.$$

Hence, using the following expansion:

$$\ln(1 + hx) = hx - \frac{(hx)^2}{2} + \frac{(hx)^3}{3} - \frac{(hx)^4}{4} + \dots + (-1)^{n+1} \frac{(hx)^n}{n},$$

we arrive at:

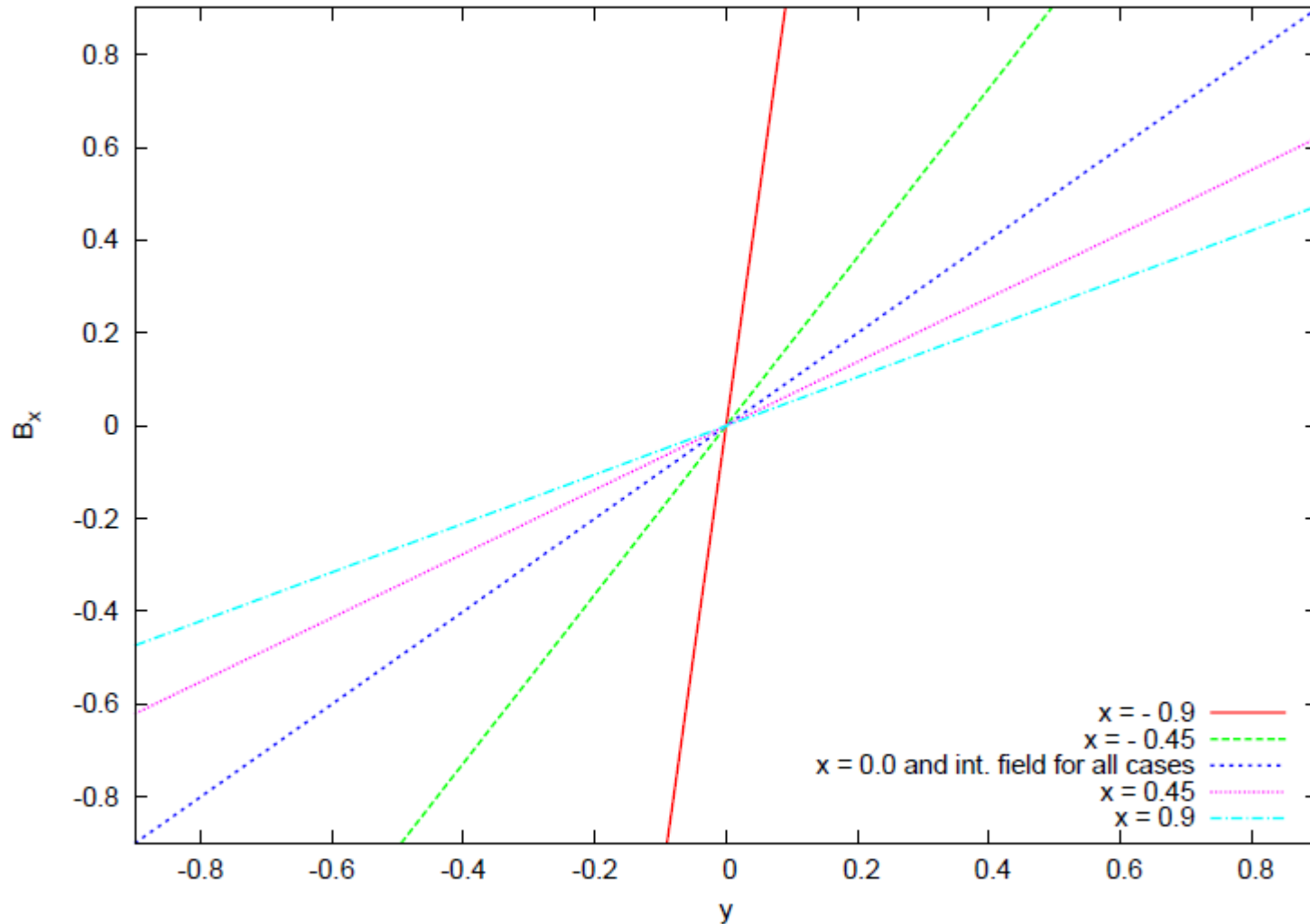
$$g = \frac{c_1 d_2}{h} \left[ \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(hx)^n}{n} \right] = c_1 d_2 x + c_3 + \dots$$

So, for a combined function magnet consisting of just dipole and quadrupole, we can write:

$$B_x = \frac{c_1 d_2 y}{(1 + hx)}, \quad B_y = \frac{c_1 d_2}{h} \ln(1 + hx) + c_3, \quad B_s = 0.$$

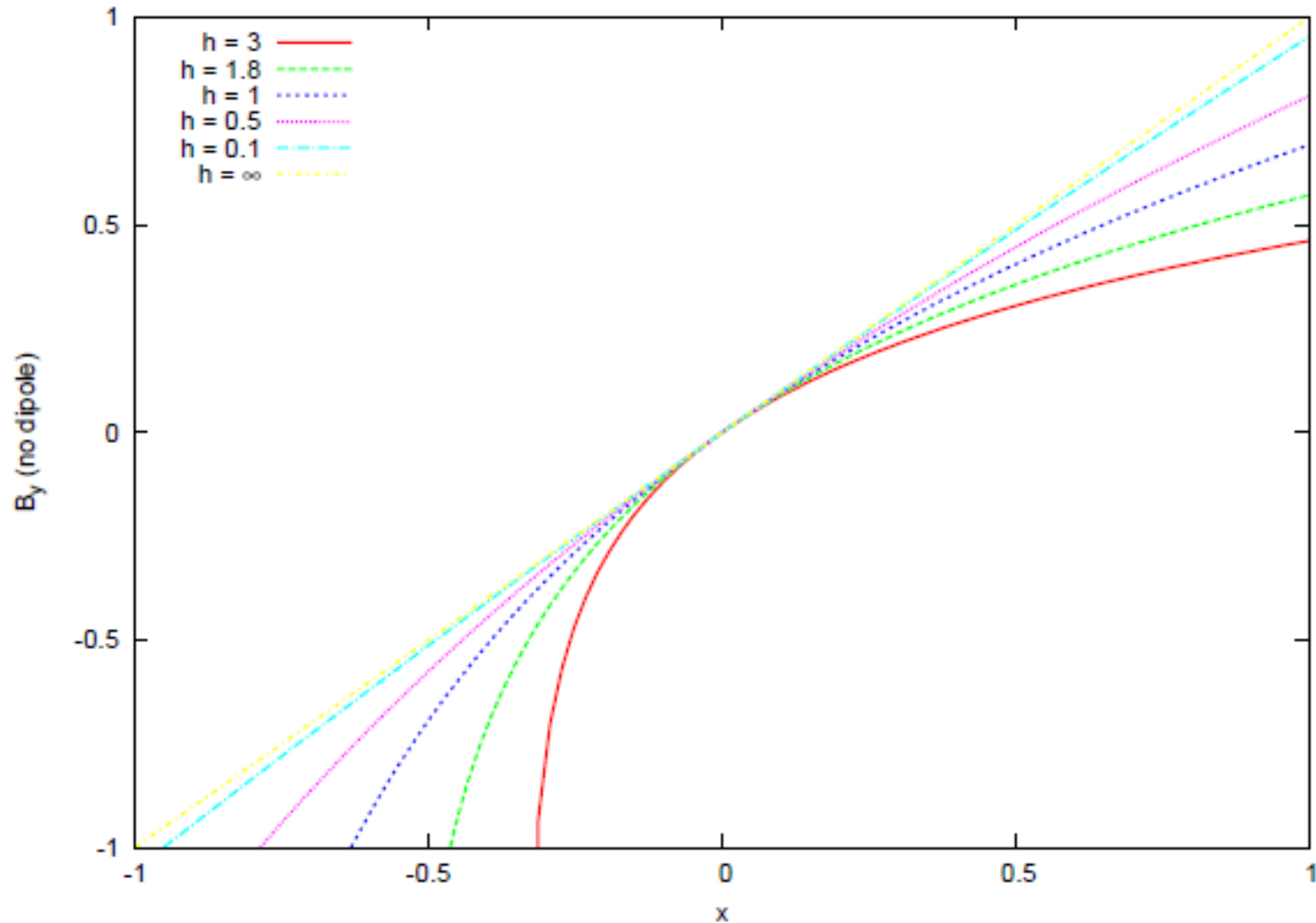
with  $c_3 = B_0$ , and  $c_1 d_2 = B_1$ . Plotting  $B_x$  and  $B_y$  we have:

# CFM – dipole & quadrupole (3)



This appears to make sense because the field,  $B_x$ , gets stronger as we go from the middle of the combined function magnet towards its centre.

# CFM – dipole & quadrupole (4)



Similarly,  $B_y$  behaves as expected (I think) as  $h$ , or the inverse of the radius of curvature varies. Note that  $B_0$ , the dipole component has been subtracted.

# “Intelligent” magnet design (1)

- Idea: put together the pole-face work with the combined function magnet work to be able to come up with ideal magnet design (ongoing with Kiril Marinov, Andy Wolski, student).
- This would give control of fringe fields & how they decay & which multipole orders are present in the magnet
- Use full Enge model or alternative so as to be able to describe the desired magnet fall-off with maximum accuracy
- Make the magnet easier to design ideally - ?
- Make the magnet easier to build
- Implement higher order multipoles into GPT – useful for other space charge codes ?
- Implement combined function magnets into GPT
- Look at potentials to develop symplectic kicks
- Anything else ?

# Conclusions

- Works for multipoles, not just quadrupoles with analytical formula for each case as long as they are built out of the correct functions ( $f(h+iz)$  and  $g(h-iz)$ )
- Decay does not have to be Enge-type. There are others (also singular) but the only real difference would be the rapidity of the fall-off
- $B_z$  is a relevant quantity which seems to be often ignored and can be as much as 1/10 or more of the other components (depending on the value of  $b$  and considering the region where field is maximum & non-singular)
- Also useful for the inner triplet region for HL-LHC because beam goes through this quite a lot off-axis so the fringe fields of the quadrupoles may matter considerably.
- This was only done for the simplest Enge-type fall off and to give an idea what the fields are and it now has to be repeated for a more general case with  $(a_0, a_1, a_2, a_3, a_4, a_5)$  non-zero.
- Having the 3 vector potential allows for symplectic expressions to be derived for the 'kick' due to any multipole fringe field hopefully following a recipe of Y.K. Wu, E. Forest and D.S. Robin which may be useful for implementation in tracking programs.
- Could it quicken certain aspects of magnet design ?
- Extension to any combined function magnet straightforward ...

**Thank you very much for your time**

**Any questions / comments / suggestions ?**